Carl Friedrich Gauss

Collected Works

Volume III

A New Proof of the Theorem That Every Polynomial in One Variable Can Be Resolved into Real Factors of the First or Second Degree

Complete works of 1799, 1815, 1816, 1828 and 1849

Modern English translation

The historical title matter and the complete 1799 dissertation, Articles 1–24, follow, with Gauss's ten historical footnotes and all four figures. The complete second proof, Articles 1–20 with both historical footnotes, follows with its title matter. It was presented in 1815 and published in 1816. The complete third proof, presented on 30 January 1816 and published that year, follows with its title matter, introduction, Articles 1–4 and historical footnote. The complete 1828 paper Proof of an Algebraic Theorem follows, on positive and negative roots and changes and continuations of coefficient signs, including missing terms. The complete Contributions to the Theory of Algebraic Equations, read on 16 July 1849 and published in 1850, follow with their title, introduction, Articles 1–20, two historical footnotes and figure. Later Volume-III works are outside this reader. Historical references retain the source numbering.

Historical source: Carl Friedrich Gauss, *Werke*, Volume III, Göttingen, 1866; dissertation first published at Helmstedt by C. G. Fleckeisen, 1799. The original German title matter and the papers of 1828 and 1849 are rendered in modern German in the German reader; their historical originals are retained separately.

AI-translated into English using OpenAI GPT-6.1 Sol; AI-produced typesetting.

Translation sources and terminology

Historical text: Carl Friedrich Gauss, *Werke*, Volume III, Göttingen, 1866. These English and German texts are AI-translated from the original-language edition. The following human translations and writings are consulted for the stated terminology and usage; their running text is not reproduced. Gauss's original mathematical formulations remain controlling.

Paul Taylor; Bernard Leak. Taylor's English translation of Gauss's second proof of the fundamental theorem of algebra, with corrections by Leak, 1983. The corresponding passages of the 1815 second proof were consulted for English polynomial, divisor, degree, symmetric-function and proof terminology. Related terminology was also consulted for the 1799 dissertation, the 1828 sign-rule paper and the 1849 introduction and Articles 1 and 10; this is not a parallel translation of those works. No running text from this human translation is adopted. Its later use of complex quantities does not replace Gauss's definition of imaginary quantities here. The consulted online edition identifies the translation as December 1983, its LaTeX version as August 2003, and the underlying source as last modified on 8 June 2007. Its acknowledgements credit Mark Wainwright for returning Taylor's notes and Nikita Danilov for typo corrections.
`https://www.paultaylor.eu/misc/gauss.html`

H. Maser. German translator and editor, *Carl Friedrich Gauss' Untersuchungen über höhere Arithmetik*, Berlin: Julius Springer, 1889. Related German terminology for algebraic functions, roots, divisibility, quotients, degrees, logarithms and tables; not a parallel translation of these works.
`https://archive.org/details/carlfriedrichga00gausgoog`

James Caddall Morehead and Adam Miller Hiltebeitel. Translators of Gauss's *General Investigations of Curved Surfaces of 1827 and 1825*, Princeton University Library, 1902; Project Gutenberg 36856, TeX edition. Related English geometrical, trigonometric, series, coordinate, continuity, integration, signed-area and infinitely small quantity terminology; not a parallel translation of these algebraic proofs.
`https://www.gutenberg.org/ebooks/36856`

Ulrich Görtz. *Algebra*, Wintersemester 2021/22, section 3.1.1, Examples 3.1–3.2. Related contemporary German usage for polynomials, indeterminates, sums, numerators and denominators, and mathematical argument. This is a terminology reference, not a parallel translation of these proofs. No running text or modern ring-theoretic argument is adopted.
`https://math.ug/a-ws2122/sect0006.html`

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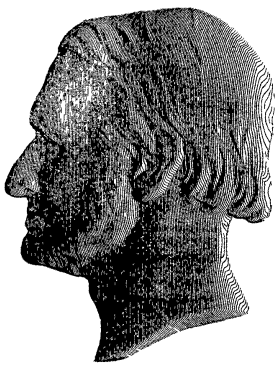
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CARL FRIEDRICH GAUSS: WORKS

VOLUME III.

CARL FRIEDRICH GAUSS
WORKS

THIRD VOLUME



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1866.

A NEW PROOF
OF THE THEOREM
THAT EVERY POLYNOMIAL

IN ONE VARIABLE
CAN BE RESOLVED INTO REAL FACTORS
OF THE FIRST OR SECOND DEGREE

SUBMITTED FOR
THE HIGHEST HONOURS IN PHILOSOPHY
TO THE ILLUSTRIOUS FACULTY OF PHILOSOPHY
OF THE ACADEMIA JULIA CAROLINA

BY

CARL FRIEDRICH GAUSS

HELMSTEDT
AT C. G. FLECKEISEN. MDCCLXXXXIX.

A NEW PROOF OF THE THEOREM
THAT EVERY POLYNOMIAL
IN ONE VARIABLE
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1.

Every determinate algebraic equation can be reduced to the form

$$x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.} + M = 0$$

where m is a positive integer. If we denote the left-hand side of this equation by X and suppose that the equation $X = 0$ is satisfied by several unequal values of x, say by setting $x = \alpha$, $x = \beta$, $x = \gamma$, etc., then the function X will be divisible by the product of the factors $x - \alpha$, $x - \beta$, $x - \gamma$, etc. Conversely, if the product of several linear factors $x - \alpha$, $x - \beta$, $x - \gamma$, etc. divides the function X, then the equation $X = 0$ will be satisfied by setting x equal to any of the quantities α, β, γ, etc. Finally, if X equals a product of m such linear factors (whether they are all different or some of them are identical), no linear factors other than these can divide the function X. An equation of degree m therefore cannot have more than m roots. At the same time, however, it is evident that an equation of degree m can have *fewer* roots even if X can be resolved into m linear factors: for if some of these factors are identical, the number of different ways of satisfying the equation will necessarily be smaller than m. Yet, for the sake of uniformity, mathematicians have preferred to say that the equation has m roots in this case too, with only some of them being equal to one another; which they may certainly allow themselves to do.

2.

What has been set out so far is adequately proved in books on algebra and nowhere offends geometric rigour. But analysts seem to have adopted too hastily, and without a prior sound proof, the theorem on which almost the entire theory of equations rests: *every function such as X can always be resolved into m linear factors*, or, what agrees with this completely, *every equation of degree m actually has m roots*. Since even quadratic equations very often lead to cases that contradict this theorem, algebraists, in order to bring these cases under it, were forced to invent an imaginary quantity whose square is -1. They then recognized that, if quantities of the form $a + b\sqrt{-1}$ are admitted in the same way as real quantities, the theorem holds not only for quadratic equations but also for cubic and biquadratic equations.

Yet this by no means justified concluding that, once quantities of the form $a + b\sqrt{-1}$ are admitted, every equation of the fifth or a higher degree can be satisfied, or, as it is usually expressed (though I would not favour this slippery wording), that the roots of every equation can be reduced to the form $a+b\sqrt{-1}$. In substance, this theorem differs in no respect from the one stated in the title of this work; giving a new rigorous proof of it is the aim of the present dissertation.

Moreover, ever since analysts discovered that there are infinitely many equations that would have no root at all unless quantities of the form $a+b\sqrt{-1}$ were admitted, such fictitious quantities have been regarded as a special kind of quantity, called *imaginary* to distinguish them from *real* quantities, and have been introduced throughout analysis. By what right? I do not discuss that here. — I shall complete my proof without any help from imaginary quantities, although I too would be entitled to use the same freedom that all recent analysts have used.

3.

Although what most elementary books offer as a proof of our theorem is so flimsy and so far removed from geometric rigour as scarcely to deserve mention, I shall nevertheless touch on it briefly so that nothing is left out. 'To show that every equation

$$x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.} + M = 0$$

or $X = 0$ actually has m roots, they undertake to prove that X can be resolved into m linear factors. For this purpose they assume m linear factors $x - \alpha$, $x - \beta$, $x - \gamma$, etc., where α, β, γ, etc. are still unknown, and set their product equal to the function X. Then, by comparing coefficients, they obtain m equations from which, they say, the unknowns α, β, γ, etc. can be determined, since their number is also m. Namely, $m - 1$ unknowns can be eliminated, leaving an equation containing only whichever unknown one wishes.' Passing over the other objections that could be made to such reasoning, I ask only: how can we be certain that this final equation actually has any root? Why could it not happen that no magnitude in the entire range of real and imaginary quantities satisfies either this final equation or the proposed one? — Experts will, moreover, readily see that this final equation must be *entirely identical* with the proposed one if the calculation has been carried out correctly: namely, eliminating the unknowns β, γ, etc. must give the equation

$$\alpha^m + A\alpha^{m-1} + B\alpha^{m-2} + \text{etc.} + M = 0$$

There is no need to say more about this reasoning.

Some authors who seem to have noticed the weakness of this method assume as an *axiom* that every equation actually has roots: if not possible roots, then impossible ones. They do not seem to have made sufficiently clear what they mean by possible and impossible quantities. If possible quantities are to mean the same as real quantities, and impossible quantities the same as imaginary ones, that axiom can by no means be admitted: it necessarily requires a proof. Yet the expressions do not seem intended in that sense; rather, the axiom seems to mean this: 'Although we are not yet certain that there must be m real or imaginary quantities satisfying a given equation of degree m, we shall nevertheless suppose this for a while; for if it should happen that so many real and imaginary quantities cannot be found, we shall certainly have a way out by saying that the remaining ones are impossible.' If someone prefers this wording to saying simply that the equation would not have so many roots in this case, I have no objection. But if he then uses these impossible roots as though they were something real, and says, for example, that the sum of all the roots of the equation $x^m + Ax^{m-1} + \text{etc.} = 0$ is $= -A$, even if there are impossible roots among them (an expression that properly means *even if some are missing*), I cannot approve of this at all. For impossible roots, understood in this sense, are still roots; that axiom therefore cannot possibly be admitted without proof, and would it be unreasonable to wonder whether there might be equations that do not even have impossible roots?[*)]

[*)] Here I always understand an imaginary quantity to mean a quantity of the form $a+b\sqrt{-1}$, as long as b is not $= 0$. All mathematicians of the first rank have always understood the expression in this sense. I do not think we should listen to those who have wished to call a quantity $a + b\sqrt{-1}$ imaginary only when $a = 0$, but impossible when $a = 0$ is not the case, since this distinction is neither necessary nor of any use. — If imaginary quantities are to be retained in analysis at all (which, for several reasons, seems more advisable than abolishing them, provided they are established on a sufficiently sound basis), they must necessarily be regarded as just as possible as real quantities. I would therefore prefer to include real and imaginary quantities under the common name *possible quantities*. In contrast, I would call a quantity *impossible* if it were required to satisfy conditions that cannot be satisfied even when imaginary quantities are admitted, but in such a way that this *wording* means the same as saying that no such quantity exists anywhere in the range of magnitudes. I would by no means allow a special kind of quantity to be formed from this. If someone says that a rectilinear equilateral right-angled triangle is impossible, no one will deny it. But if he wished to regard such an impossible triangle as a new kind of triangle and apply other properties of triangles to it, who could keep from laughing? This would be playing with words, or rather abusing them. — Even the greatest mathematicians have quite often applied truths that clearly presuppose the possibility of the quantities concerned to quantities whose possibility was still doubtful. Nor would I deny that such liberties generally concern only the form and, as it were, the covering of the reasoning, which a true mathematician's sharp eye can soon penetrate. Yet it seems more advisable, and more worthy of the eminence of a science rightly celebrated as the most perfect model of clarity and certainty, either to banish such freedoms altogether, or at least to use them more sparingly and only where even the less practised can see that the matter could have been completed without their help, perhaps less briefly, but just as rigorously. — I would not, moreover, deny that what I have said here against the misuse of

4.

Before reviewing other mathematicians' proofs of our theorem and setting out what seems objectionable to me in each, I note that it is enough to show only that every equation of any degree

$$x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.} + M = 0$$

or $X = 0$ (where the coefficients A, B, etc. are supposed real) can be satisfied in at least one way by a value of x of the form $a + b\sqrt{-1}$. For it is known that X will then be divisible by the real quadratic factor $xx - 2ax + aa + bb$ if b is not $= 0$, and by the real linear factor $x - a$ if $b = 0$. In either case the quotient will be real and of lower degree than X; and since, by the same reasoning, this quotient must have a real factor of the first or second degree, it is evident that continuing this operation will ultimately resolve the function X into real linear or quadratic factors, or, if one prefers to use two imaginary linear factors in place of each real quadratic factor, into m linear factors.

5.

The first proof of the theorem is due to the distinguished mathematician D'Alembert, *Recherches sur le calcul intégral, Histoire de l'Acad. de Berlin, Année* 1746, p. 182 ff. It is also found in Bougainville, *Traité du calcul intégral, à Paris* 1754, p. 47 ff. The principal steps of this method are as follows.

First he shows that, if any function X of the variable x becomes $= 0$ either for $x = 0$ or for $x = \infty$, and can attain an infinitely small positive real value by assigning a real value to x, this function can also attain an infinitely small negative real value through a value of x that is either real or of the imaginary form $p + q\sqrt{-1}$. Namely, if Ω denotes an infinitely small value of X, and ω the corresponding value of x, he asserts that ω can be expressed by a rapidly convergent series $a\Omega^\alpha + b\Omega^\beta + c\Omega^\gamma$, etc., where the exponents α, β, γ, etc. are continually increasing rational quantities, which therefore, at least beyond some fixed distance from the beginning, become positive and make the terms in which they occur infinitely small. Now, if none of these exponents is a fraction with an even denominator, all the terms of the series become real for both positive and negative values of Ω. If, however, there are fractions with even denominators among these exponents, then for a negative value of Ω the corresponding terms are known to have the form $p + q\sqrt{-1}$. But because of the infinite convergence of the series, in the former case it is enough to retain only the first (that is, the largest) term; in the latter there is no need to proceed beyond the term that first produces an imaginary part.

impossible quantities could in a certain respect also be objected against imaginary quantities; but I reserve their defence, together with a fuller account of this entire matter, for another occasion.

By similar reasoning it can be shown that, if X can attain an infinitely small negative value from a real value of x, this function can attain an infinitely small positive real value from a real value of x or an imaginary one of the form $p + q\sqrt{-1}$.

From this he concludes, secondly, that there is also some finite real value of X, negative in the former case and positive in the latter, that can be produced by an imaginary value of x of the form $p + q\sqrt{-1}$.

It follows that, if X is a function of x that attains a real value V from a real value v of x, and also attains from a real value of x a real value greater or smaller by an infinitely small quantity, it can also take a real value smaller or greater, respectively, than V by an infinitely small and hence by a finite quantity, by assigning to x a value of the form $p + q\sqrt{-1}$. This is readily derived from what precedes by conceiving $V + Y$ substituted for X, and $v + y$ for x.

Finally, the distinguished D'ALEMBERT asserts that, if X is supposed capable of traversing an entire interval between two real values R, S (that is, of becoming equal both to R and to S, and to all intermediate real values), by assigning to x values always of the form $p + q\sqrt{-1}$, the function X can be increased or decreased still further by any finite real quantity (according as $S > R$ or $S < R$), while x always remains of the form $p + q\sqrt{-1}$. For if there were a real quantity U (with S supposed to lie between it and R) to which X could not become equal through such a value of x, there must be a *maximum* value of X (when $S > R$; a minimum when $S < R$), say T, attained from a value $p + q\sqrt{-1}$ of x, such that no value of the same form could be assigned to x that would bring the function X any closer to U, even by the smallest amount. Now, if in this equation between X and x we substitute $p + q\sqrt{-1}$ everywhere for x and then set equal to zero both the real part and the part containing the factor $\sqrt{-1}$, after omitting that factor, two other equations can be obtained by elimination from the two resulting equations (which contain p, q and X together with constants). One contains p, X and constants; the other, free of p, contains only q, X and constants. Thus, since X has traversed all values from R to T through real values of p, q, what precedes shows that X can approach U still more closely by assigning to p, q values of the forms $\alpha + \gamma\sqrt{-1}$, $\beta + \delta\sqrt{-1}$, respectively. But then $x = \alpha - \delta + (\gamma + \beta)\sqrt{-1}$, that is, it is still of the form $p + q\sqrt{-1}$, contrary to the hypothesis.

Now, if X is supposed to denote a function such as $x^m + Ax^{m-1} + Bx^{m-2} +$ etc. $+ M$, it is readily seen that real values can be assigned to x such that X traverses an entire interval between two real values. Hence x can attain some value of the form $p + q\sqrt{-1}$ from which X becomes $= 0$. Q. E. D.*)

*) It should be noted that, in his own account of this proof, the distinguished D'ALEMBERT used geometrical considerations, regarding X as the abscissa and x as the ordinate of a curve (following the practice of all mathematicians in the first part of this century, among whom

6.

The objections that seem possible against D'ALEMBERT's proof come down chiefly to the following.

1. The distinguished d'A. raises no doubt about the *existence* of values of x corresponding to given values of X, but assumes it and investigates only the *form* of those values.

Although this objection is in itself very serious, here it concerns only the form of expression, which can easily be corrected so as to remove it entirely.

2. The assertion that ω can always be expressed by a series of the kind he gives is certainly false if X is also to designate any transcendental function (as d'A. suggests in several places). This is evident, for example, if we put $X = e^{\frac{1}{x}}$, or $x = \frac{1}{\log X}$. Yet if we restrict the proof to the case in which X is an algebraic function of x (which suffices for the present purpose), the proposition is indeed true. — d'A., moreover, has offered nothing to support his supposition. The celebrated BOUGAINVILLE supposes X to be an algebraic function of x and recommends NEWTON's parallelogram for finding the series.

3. He uses infinitely small quantities more freely than is consistent with geometric rigour, or than a careful analyst would allow, at least in our own time (when they have a deservedly bad reputation). Nor does he make the leap from an infinitely small value of Ω to a finite one sufficiently clear. He seems to infer his proposition that Ω can also attain some finite value, less from the possibility of an infinitely small value of Ω than from this: if Ω denotes a very small quantity, then because of the rapid convergence of the series, the more terms of the series are taken, the closer one comes to the true value of ω; or, the more terms are included in the sum taken for ω, the more accurately the equation expressing the relation between ω and Ω, or x and X, will be satisfied. Apart from the fact that this entire argument seems too vague to yield any rigorous conclusion, I note that there are indeed series that, however small a value is assigned to the quantity in whose powers they proceed, nevertheless always diverge, so that, if only they are continued far enough, one can reach terms greater than any given quantity[*)]. This happens

the notion of functions was less familiar). Since, however, all his reasoning, considered only in its substance, rests on no geometrical principles but on purely analytical ones, and the expressions imaginary curves and imaginary ordinates seem rather harsh and more liable to offend the present-day reader, I have preferred a purely analytical presentation here. I added this note so that no one comparing D'ALEMBERT's own proof with this brief account should suspect that anything essential has been changed.

[*)] I note in passing on this occasion that many of these series seem at first sight extremely convergent: for example, most of those that the distinguished EULER uses in the latter part of the *Inst. Calc. Diff.*, Chapter VI, pp. 441–474, to assign as closely as possible the sums of other series (the remaining series on pp. 475–478 can indeed converge). As far as I know, no one has observed this before. It would therefore be very desirable to show clearly and rigorously why series of this kind, which at first converge very rapidly, then gradually more

when the coefficients of the series form a hypergeometric progression. It was therefore necessary to prove that such a hypergeometric series could not arise in the present case.

Moreover, it seems to me that the distinguished d'A. was wrong to resort to infinite series here, and that they are unsuitable for establishing this fundamental theorem of the theory of equations.

4. From the supposition that X can attain the value S, but not the value U, it does not yet follow that between S and U there must lie a value T that X can reach but cannot exceed. Another case still remains: there could be a limit between S and U that X can approach as closely as one wishes, yet never reach. From the arguments given by the distinguished d'A. it follows only that X can exceed every value it has reached by a further finite quantity. For instance, once it has become $= S$, it can be increased by some further finite quantity Ω; after this a new increment Ω' can be added, then another increment Ω'', etc., so that however many increments have already been added, none should be regarded as the last, but a new one can always be added. Yet although the *number* of possible increments is bounded by no limits, it could certainly happen that, if the increments $\Omega, \Omega', \Omega''$, etc. continually decreased, the sum $S + \Omega + \Omega' + \Omega''$, etc. would nevertheless never attain some limit, however many terms were considered.

Although this case cannot occur when X denotes a polynomial in x, without a proof that it cannot happen the method must nevertheless be regarded as incomplete. When, however, X is a transcendental function, or even an algebraic fractional function, that case can indeed occur: for example, whenever an infinitely large value of x corresponds to some value of X. Then D'ALEMBERT's method seems incapable of being reduced to unquestionable principles without many complications, and perhaps in some cases not at all.

For these reasons I cannot regard D'ALEMBERT's proof as satisfactory. Nevertheless, it seems to me that its real force is by no means broken by all these objections. I believe that on the same foundation (though in a very different way, and at least with greater circumspection) not only can a rigorous proof of our theorem be built, but everything one could wish for in the theory of *transcendental* equations can be obtained from it. I shall discuss this very important matter more fully on another occasion; meanwhile, compare Article 24 below.

and more slowly, and finally diverge more and more, nevertheless give a nearly true sum provided not too many terms are taken, and how far such a sum can safely be regarded as exact?

7.

After D'ALEMBERT, the distinguished EULER published his investigations on the same subject in *Recherches sur les racines imaginaires des équations, Hist. de l'Acad. de Berlin A.* 1749, p. 223 ff. He gave two methods there; the substance of the first is as follows.

First the distinguished Euler undertakes to prove that, if m denotes any power of the number 2, the function $x^{2m} + Bx^{2m-2} + Cx^{2m-3} + \text{etc.} + M = X$ (in which the coefficient of the second term is $= 0$) can always be resolved into two real factors in which x rises to degree m. For this purpose he assumes two factors,

$$x^m - ux^{m-1} + \alpha x^{m-2} + \beta x^{m-3} + \text{etc.}, \quad \text{and} \quad x^m + ux^{m-1} + \lambda x^{m-2} + \mu x^{m-3} \text{ etc.}$$

where the coefficients u, α, β, etc., λ, μ, etc. are still unknown, and sets their product equal to the function X. Comparing coefficients then gives $2m - 1$ equations, and evidently all that must be proved is that real values satisfying these equations can be assigned to the unknowns u, α, β, etc., λ, μ, etc. (whose number is also $2m-1$). Euler now asserts that, if u is first regarded as known, so that the number of unknowns is one less than the number of equations, then by suitably combining the equations according to known algebraic methods, all of α, β, etc., λ, μ, etc. can be determined rationally, without any extraction of roots, from u and the coefficients B, C, etc., and thus attain real values as soon as u becomes real. Moreover, all of α, β, etc., λ, μ, etc. can be eliminated to give an equation $U = 0$, where U is a polynomial in u alone and the known coefficients. Expanding this equation itself by the usual method of elimination would be an immense task when the proposed equation $X = 0$ is of a rather high degree, and completely impossible for an unspecified degree (according to Euler himself, p. 239). Yet here it is enough to know one property of this equation: the constant term in U (which does not involve the unknown u) is necessarily negative. From this it is known to follow that the equation has at least one real root, or that u, and therefore also α, β, etc., λ, μ, etc., can be determined as real quantities in at least one way. That property can be established by the following considerations. Since $x^m - ux^{m-1} + \alpha x^{m-2} + \text{etc.}$ is supposed to be a factor of the function X, u must be the sum of m roots of the equation $X = 0$, and must therefore have as many values as there are different ways of choosing m out of $2m$ roots; that is, by the principles of combinatorics, $\frac{2m.2m-1.2m-2.\ldots.m+1}{1.2.3\ldots m}$ values. This number is always twice an odd number (I omit the proof, which is not difficult); if it is put $= 2k$, its half k is therefore odd, while the equation $U = 0$ has degree $2k$. Now, since the second term is absent in the equation $X = 0$, the sum of all $2m$ roots is 0. Thus, if the sum of any m roots is $+p$, the sum of the remaining ones is $-p$: that is, if $+p$ is among the values of u, then $-p$ is among them too. From this Euler concludes that U is the product of k quadratic factors such

as $uu - pp$, $uu - qq$, $uu - rr$, etc., where $+p$, $-p$, $+q$, $-q$, etc. denote all $2k$ roots of the equation $U = 0$. Because the number of these factors is odd, the constant term in U is therefore the square of the product pqr, etc., with a negative sign. But the product pqr, etc. can always be determined rationally from the coefficients B, C, etc., and must therefore be a real quantity. Its square with a negative sign is thus certainly a negative quantity. Q. E. D.

Since these two real factors of X have degree m, and m is a power of the number 2, each can in the same way be resolved again into two real factors of degree $\frac{1}{2}m$. Repeated halving of the number m must eventually lead to two; hence continuing the operation evidently gives the function X resolved into real quadratic factors.

If, however, a function is proposed in which the second term is not absent, say $x^{2m} + Ax^{2m-1} + Bx^{2m-2} + \text{etc.} + M$, where $2m$ still denotes a power of two, the substitution $x = y - \frac{A}{2m}$ transforms it into a similar function without a second term. It readily follows that this function too can be resolved into real quadratic factors.

Finally, given a function of degree n, where n denotes a number that is not a power of two, let the next larger power of two than n be put $= 2m$, and multiply the proposed function by any $2m - n$ real linear factors. From the fact that the product can be resolved into real quadratic factors, it readily follows that the proposed function too must be resolvable into real factors of the second or first degree.

8.

The following objections can be made to this proof.

1. The rule by which Euler concludes that all $2m - 2$ unknowns α, β, etc., λ, μ, etc. can be determined rationally from $2m - 1$ equations is by no means general, but very often admits exceptions. If, for example, in Article 3 one regards one of the unknowns as known and tries to express the others rationally in terms of it and the given coefficients, one readily finds that this is impossible, and that no unknown quantity can be determined except by an equation of degree $m - 1$. Although here one can immediately see a priori that this must necessarily happen, one might nevertheless reasonably doubt whether, in the present case too, matters stand in the same way for some values of m, so that the unknowns α, β, etc., λ, μ, etc. cannot be determined from u, B, C, etc. except by an equation of degree perhaps higher than $2m$. For the case where the equation $X = 0$ is of the fourth degree, Euler derives rational values of the coefficients in terms of u and the given coefficients; but that this can also be done for all higher equations certainly needed a fuller explanation. — It seems worthwhile, moreover, to investigate more deeply and in the greatest generality those formulas that express α, β, etc. rationally in terms of u, B, C, etc. I shall undertake a fuller treatment of this

and other matters belonging to the theory of elimination (a subject by no means exhausted) on another occasion.

2. Even if it has been proved that, whatever the degree of the equation $X = 0$, formulas can always be found that express α, β, etc., λ, μ, etc. rationally in terms of u, B, C, etc., it is nevertheless certain that for some particular values of the coefficients B, C, etc. these formulas can become *indeterminate*, so that not only is it impossible to determine these unknowns rationally from u, B, C, etc., but in some cases there are actually no real values of α, β, etc., λ, μ, etc. corresponding to a real value of u. To confirm this, for brevity I refer the reader to Euler's paper itself, where the equation of the fourth degree is discussed more fully on p. 236. Everyone will immediately see that the formulas for the coefficients α, β become indeterminate if $C = 0$ and the value 0 is taken for u, and that their values not only cannot be assigned without extracting roots, but are not even real if $BB - 4D$ is a negative quantity. Although in this case one can readily see that u has other real values to which real values of α, β correspond, one might nevertheless fear that unraveling this difficulty (which Euler did not touch on at all) could require much greater effort in higher equations. Certainly this matter must by no means be passed over in silence in an exact proof.

3. The distinguished Euler tacitly supposes that the equation $X = 0$ has $2m$ roots, and puts their sum $= 0$ because the second term is absent in X. I have already stated above, in Article 3, what I think of this freedom (which all authors on this subject use). The proposition that the sum of all the roots of an equation equals its first coefficient with its sign changed seems applicable only to equations that have roots. Since this very proof is meant to establish that the equation $X = 0$ actually has roots, it seems inadmissible to suppose their existence. No doubt those who have not yet seen through the fallacy in this reasoning will reply that *what is proved here is not that the equation* $X = 0$ *can be satisfied* (for that is what the expression that it has roots means), *but only that it can be satisfied by values of* x *contained in the form* $a + b\sqrt{-1}$*; the former is supposed as an axiom.* But since no forms of quantities other than the real and the imaginary $a + b\sqrt{-1}$ can be conceived, it is not sufficiently clear how what is to be proved differs from what is supposed as an axiom. Indeed, even if it were possible to devise still other forms of quantities, say the form F, F', F'', etc., it should nevertheless not be admitted without proof that an equation can be satisfied by some value of x that is either real, or contained in the form $a + b\sqrt{-1}$, or in the form F, or in F', etc. That axiom can therefore have no other meaning than this: any equation can be satisfied *either* by a real value of the unknown, *or* by an imaginary value contained in the form $a+b\sqrt{-1}$, *or* perhaps by a value contained in some other form hitherto unknown, *or* by a value contained in no form at all. But how quantities of this kind, of which you cannot even

form an idea — a true shadow of a shadow — can be added or multiplied is certainly not understood with the clarity always required in mathematics*).

I do not, however, wish these objections to cast any doubt on the conclusions Euler drew from his supposition. On the contrary, I am certain that they can be established by a method neither difficult nor very different from Euler's, in such a way that not even the slightest doubt should remain for anyone. I criticize only the *form*, which, although it can be very useful in *discovering* new truths, nevertheless seems quite unacceptable in *proving* them before the public.

4. The distinguished Euler offered nothing at all to prove the assertion that the product pqr, etc. can be determined *rationally* from the coefficients in X. Everything he says about this for equations of the *fourth degree* is as follows (where a, b, c, d are the roots of the proposed equation $x^4+Bxx+Cx+D = 0$):

'No doubt it will be objected that I have supposed here that the quantity pqr was a real quantity and that its square $ppqqrr$ was positive; this was still doubtful, since, the roots a, b, c, d being imaginary, it could well happen that the square of the quantity pqr formed from them was negative. To this I reply that this case can never occur; for however imaginary the roots a, b, c, d may be, we nevertheless know that there must be $a + b + c + d = 0$: $ab + ac + ad + bc + bd + cd = B$; $abc + abd + acd + bcd = -C$*); $abcd = D$, these quantities B, C, D being real. But since $p = a + b$, $q = a + c$, $r = a + d$, their product $pqr = (a + b)(a + c)(a + d)$ can be determined, *as we know*, from the quantities B, C, D, and will consequently be real, just as we have seen that in fact $pqr = -C$ and $ppqqrr = CC$. It will readily be recognized in the same way that this same circumstance must occur in higher equations, and that no objections can be made to me on this ground.' Euler nowhere added the condition that the product pqr, etc. can be determined *rationally* in terms of B, C, etc.; nevertheless, he seems always to have understood it, since without it the proof can have no force. Now it is indeed true for equations of the fourth degree that expanding the product $(a + b)(a + c)(a + d)$ gives $aa(a+b+c+d)+abc+abd+acd+bcd = -C$; yet it does not seem sufficiently clear how in all higher equations the product can be determined rationally from the coefficients. The celebrated de Foncenex, who first observed this (*Miscell. phil. math. soc. Taurin.*, Vol. I, p. 117), rightly maintains that without a rigorous proof of this proposition the method loses all its

*)This whole matter will be greatly clarified by another investigation already in the press, where, on a subject very different but nevertheless analogous, I could have used a similar freedom with exactly the same right as all analysts have done here with equations. Although, with the help of such fictions, I could have completed in a few words the proofs of several truths that without them became very difficult and require the most subtle devices, I nevertheless preferred to refrain from them altogether, and I hope that few would have been satisfied had I imitated the analysts' method.

*)Euler has C in error, and hence also later wrongly puts $pqr = C$.

force; he confesses, however, that this proof seems rather difficult to him, and describes the path he tried in vain*). Nevertheless, the matter can be settled without much difficulty by the following method (whose substance I can only indicate here). Although in equations of the fourth degree it is not sufficiently clear that the product $(a+b)(a+c)(a+d)$ can be determined in terms of the coefficients B, C, D, one can readily see that the same product is also $= (b+a)(b+c)(b+d)$, as well as $= (c+a)(c+b)(c+d)$, and finally also $= (d+a)(d+b)(d+c)$. The product pqr will therefore be one quarter of the sum $(a+b)(a+c)(a+d)+(b+a)(b+c)(b+d)+(c+a)(c+b)(c+d)+(d+a)(d+b)(d+c)$, which, when expanded, can readily be foreseen a priori to be a polynomial in the roots a, b, c, d in which they all enter in the same way. Such functions can always be expressed rationally in terms of the coefficients of the equation whose roots are a, b, c, d. — The same thing is also evident if the product pqr is put into the form

$$\tfrac{1}{2}(a+b-c-d) \times \tfrac{1}{2}(a+c-b-d) \times \tfrac{1}{2}(a+d-b-c)$$

for one can readily foresee that this product, when expanded, will involve all of a, b, c, d in the same way. Those familiar with the subject will also readily gather from this how it must be applied to higher equations. — A complete account of the proof, which brevity does not allow me to include here, together with a fuller investigation of functions involving several variables in the same way, I reserve for another occasion.

I observe, moreover, that besides these four objections there are still some other points in Euler's proof that could be criticized; I pass over them in silence, however, lest I seem too severe a critic, especially since the foregoing seems sufficiently to show that the proof, at least in the form Euler proposed it, can by no means be regarded as complete.

After this proof Euler showed another way of reducing the theorem for equations whose degree is not a power of two to the solution of equations of that kind. But since this method teaches nothing for equations whose degree is a power of two, and is moreover open to all the preceding objections (except the fourth) just as the first general proof is, there is no need to discuss it more fully here.

*) An error seems to have crept into this account: namely, on p. 118, line 5, instead of the symbol p ("one chose only those in which p occurred", etc.), one must necessarily read *one and the same arbitrary root of the proposed equation*, or something similar, since the former makes no sense.

9.

In the same paper, on p. 263, the distinguished Euler tried to establish our theorem in yet another way, whose substance is as follows. Given an equation $x^n + Ax^{n-1} + Bx^{n-2}$ etc. $= 0$, no analytical expression representing its roots has yet been found if the exponent $n > 4$; nevertheless it seems certain (as Euler asserts) that it can contain nothing but arithmetical operations and extractions of roots, combined more intricately the greater n is. If this is granted, Euler shows very well that, however intricately the radical signs are combined, the value of the formula can always be represented in the form $M + N\sqrt{-1}$, where M, N are real quantities.

Against this reasoning one can object that, after the efforts of so many great mathematicians, very little hope remains of ever arriving at the general solution of algebraic equations, so that it becomes more and more probable that such a solution is altogether impossible and contradictory. This should seem all the less paradoxical *since what is commonly called the solution of an equation is, strictly speaking, nothing but its reduction to pure equations.* For the solution of pure equations is not taught by this, but supposed; and if you express a root of the equation $x^m = H$ by $\sqrt[m]{H}$, you have by no means solved it, nor have you done more than if you devised some sign to denote a root of the equation $x^n + Ax^{n-1} + \text{etc.} = 0$ and put the root equal to it. It is true that pure equations far surpass all others because of the ease of finding their roots by approximation and because of the elegant connection all their roots have with one another. It is therefore by no means to be criticized that analysts have denoted their roots by a special sign. Yet from the fact that they have included this sign, just like the arithmetical signs of addition, subtraction, multiplication, division and raising to a power, under the name of *analytical expressions*, it by no means follows that a root of any equation can be represented by them. Or, putting words aside, it is supposed without sufficient reason that the solution of any equation can be reduced to the solution of pure equations. Perhaps it would not be so difficult to prove this impossibility with full rigor already for the fifth degree; I shall present my investigations on this matter more fully elsewhere. Here it is enough that the general solvability of equations, understood in that sense, is still very doubtful, and that therefore a proof whose entire force depends on that supposition carries no weight in the present state of the subject.

10.

Later, the celebrated DE FONCENEX too, having noticed a defect in EULER'S first proof (above, Article 8, objection 4) that he could not remove, tried another path and presented it on p. 120 of the paper cited*). It consists in the following.

Let an equation $Z = 0$ be proposed, where Z denotes a function of degree m in the unknown z. If m is an odd number, it is already known that this equation has a real root; if m is even, however, the celebrated Foncenex tries in the following way to prove that the equation has at least one root of the form $p + q\sqrt{-1}$. Let $m = 2^n i$, where i denotes an odd number, and suppose that $zz + uz + M$ is a divisor of the function Z. Then the individual values of u will be the sums of pairs of roots of the equation $Z = 0$, with their sign changed; hence u will have $\frac{m.m-1}{1.2} = m'$ values. If u is supposed to be determined by an equation $U = 0$ (where U denotes a polynomial in u and the known coefficients in Z), this equation will have degree m'. One readily sees that m' will be a number of the form $2^{n-1}i'$, where i' denotes an odd number. Now, unless m' is odd, suppose again that $uu + u'u + M'$ is a divisor of U; similar reasoning will show that u' is determined by an equation $U' = 0$, where U' is a function of degree $\frac{m'.m'-1}{1.2}$ in u'. Putting $\frac{m'.m'-1}{1.2} = m''$, we have that m'' is a number of the form $2^{n-2}i''$, where i'' denotes an odd number. Now, unless m'' is odd, let $u'u' + u''u' + M''$ be a divisor of the function U'; then u'' will be determined by an equation $U'' = 0$, which, if it is supposed to have degree m''', gives m''' as a number of the form $2^{n-3}i'''$. It is evident that in the sequence of equations $U = 0$, $U' = 0$, $U'' = 0$, etc., the one in position n will have odd degree and therefore a real root. For brevity we shall put $n = 3$, so that the equation $U'' = 0$ has a real root u'', for one readily sees that the same reasoning holds for any other value of n. The celebrated de Foncenex then asserts that the coefficient M'' can be determined rationally from u'' and the coefficients in U' (which are readily understood to be polynomials in the coefficients in Z), or from u'' and the coefficients in Z, and is therefore real. It follows that the roots of the equation $u'u' + u''u' + M'' = 0$ will be contained in the form $p + q\sqrt{-1}$; but they will evidently satisfy the equation $U' = 0$ as well. There will therefore be some value of u' contained in the form $p + q\sqrt{-1}$. Now the coefficient M' can, in the same way as before, be determined rationally from u' and the coefficients in Z, and will therefore also be contained in the form $p + q\sqrt{-1}$. Hence the roots of the equation $uu + u'u + M'$ will be contained in the same form, and will at the same time satisfy the equation $U = 0$; that is, this equation will have a root contained in the form $p + q\sqrt{-1}$. Finally, it follows from this in a similar way that M too

*)The second volume of the same *Miscellanea*, p. 337, contains elucidations of this paper; these do not, however, concern the present investigation, but the logarithms of negative quantities, which had been discussed in the same paper.

is contained in the same form, as is a root of the equation $zz + uz + M = 0$, which will evidently also satisfy the proposed equation $Z = 0$. Thus any equation will have at least one root of the form $p + q\sqrt{-1}$.

11.

Objections 1, 2 and 3 that I made against EULER'S first proof (Article 8) have the same force against this method, with this difference, however: the second objection, to which EULER'S proof was open only in some special cases, must affect the present one in every case. Namely, one can prove a priori that, even if a formula is given that expresses the coefficient M' rationally in terms of u' and the coefficients in Z, this formula must necessarily become indeterminate for several values of u'; similarly, the formula that expresses the coefficient M'' in terms of u'' becomes indeterminate for some values of u'', etc. This is seen most clearly if we take an equation of the fourth degree as an example. Let us therefore put $m = 4$, and let the roots of the equation $Z = 0$ be α, β, γ, δ. Then it is evident that the equation $U = 0$ will be of the sixth degree and its roots $-(\alpha+\beta)$, $-(\alpha+\gamma)$, $-(\alpha+\delta)$, $-(\beta+\gamma)$, $-(\beta+\delta)$, $-(\gamma+\delta)$. The equation $U' = 0$, however, will be of the fifteenth degree, and the values of u' will be

$$\begin{array}{ccc}
2\alpha+\beta+\gamma, & 2\alpha+\beta+\delta, & 2\alpha+\gamma+\delta, \\
2\beta+\alpha+\gamma, & 2\beta+\alpha+\delta, & 2\beta+\gamma+\delta, \\
2\gamma+\alpha+\beta, & 2\gamma+\alpha+\delta, & 2\gamma+\beta+\delta, \\
2\delta+\alpha+\beta, & 2\delta+\alpha+\gamma, & 2\delta+\beta+\gamma, \\
\alpha+\beta+\gamma+\delta, & \alpha+\beta+\gamma+\delta, & \alpha+\beta+\gamma+\delta
\end{array}$$

Now we must stop at this equation, since its degree is odd, and it will in fact have the real root $\alpha+\beta+\gamma+\delta$ (which equals the first coefficient in Z with its sign changed and will therefore be not only real but also rational if the coefficients in Z are rational). But one can readily see that, if a formula is given that expresses the value of M' rationally in terms of the corresponding value of u', it necessarily becomes indeterminate for $u' = \alpha+\beta+\gamma+\delta$. For this value will be a root of the equation $U' = 0$ *three times*, and three values of M' will correspond to it, namely $(\alpha+\beta)(\gamma+\delta)$, $(\alpha+\gamma)(\beta+\delta)$ and $(\alpha+\delta)(\beta+\gamma)$, which can all be irrational. Evidently a rational formula could in this case produce neither an irrational value of M' nor three different values. From this example one can sufficiently gather that the celebrated DE FONCENEX'S method is by no means satisfactory; if it is to be made complete in every respect, the theory of elimination must be investigated much more deeply.

12.

Finally, the distinguished La Grange treated our theorem in the paper *Sur la forme des racines imaginaires des équations, Nouv. Mém. de l'Acad. de Berlin* 1772, p. 222 ff. This great mathematician took particular pains to supply the defects in Euler's first proof, and in fact investigated especially those matters that constitute the second and fourth objections above (Article 8) so deeply that nothing further remains to be desired, except perhaps that some doubts seem to remain in the earlier investigation concerning the theory of elimination, on which this whole investigation rests. — Yet he did not touch on the third objection at all; indeed, the whole investigation is built on the supposition that any equation of degree m actually has m roots.

Thus, after carefully weighing what has been set out so far, I hope that those familiar with the subject will welcome a new proof of this most important theorem drawn from entirely different principles. I now proceed to give it.

13.

Lemma. *If m denotes any positive integer, the function* $\sin\varphi \,.\, x^m - \sin m\varphi \,.\, r^{m-1}x + \sin(m-1)\varphi \,.\, r^m$ *will be divisible by* $xx - 2\cos\varphi \,.\, rx + rr$.

Proof. For $m = 1$ this function becomes $= 0$ and is therefore divisible by any factor; for $m = 2$ the quotient becomes $\sin\varphi$, and for any greater value the quotient will be $\sin\varphi \,.\, x^{m-2} + \sin 2\varphi \,.\, rx^{m-3} + \sin 3\varphi \,.\, rrx^{m-4} +$ etc. $+ \sin(m-1)\varphi \,.\, r^{m-2}$. For one readily verifies that, when this function is multiplied by $xx - 2\cos\varphi \,.\, rx + rr$, the product equals the proposed function.

14.

Lemma. *If the quantity r and the angle φ are determined so that the equations*

$$\begin{aligned} &r^m \cos m\varphi + Ar^{m-1}\cos(m-1)\varphi + Br^{m-2}\cos(m-2)\varphi + \text{etc.} \\ &\qquad + Krr\cos 2\varphi + Lr\cos\varphi + M = 0 \quad . \quad . \quad [1] \\ &r^m \sin m\varphi + Ar^{m-1}\sin(m-1)\varphi + Br^{m-2}\sin(m-2)\varphi + \text{etc.} \\ &\qquad + Krr\sin 2\varphi + Lr\sin\varphi = 0 \quad . \quad . \quad [2] \end{aligned}$$

hold, the polynomial $x^m + Ax^{m-1} + Bx^{m-2} + etc. + Kxx + Lx + M = X$ will be divisible by the quadratic factor $xx - 2\cos\varphi \,.\, rx + rr$, provided we do not have $r\sin\varphi = 0$; but if $r\sin\varphi = 0$, the same polynomial will be divisible by the linear factor $x - r\cos\varphi$.

Proof. I. By the preceding article, all the following quantities will be divisible by $xx - 2\cos\varphi \,.\, rx + rr$:

$$\begin{array}{rll}
\sin\varphi \,.\, rx^m & -\sin m\varphi \,.\, r^m x & +\sin(m-1)\varphi \,.\, r^{m+1} \\
A\sin\varphi \,.\, rx^{m-1} & -A\sin(m-1)\varphi \,.\, r^{m-1}x & +A\sin(m-2)\varphi \,.\, r^m \\
B\sin\varphi \,.\, rx^{m-2} & -B\sin(m-2)\varphi \,.\, r^{m-2}x & +B\sin(m-3)\varphi \,.\, r^{m-1} \\
 & \text{etc.} & \text{etc.} \\
K\sin\varphi \,.\, rxx & -K\sin 2\varphi \,.\, rrx & +K\sin\varphi \,.\, r^3 \\
L\sin\varphi \,.\, rx & -L\sin\varphi \,.\, rx & * \\
M\sin\varphi \,.\, r & * & +M\sin(-\varphi) \,.\, r
\end{array}$$

Their sum will therefore also be divisible by $xx - 2\cos\varphi \,.\, rx + rr$. Now the first parts of these quantities form the sum $\sin\varphi \,.\, rX$; the second parts, added together, give 0 by [2]; and it is easy to see that the sum of the third parts also vanishes if we multiply [1] by $\sin\varphi$ and [2] by $\cos\varphi$, and subtract the former product from the latter. It follows that the polynomial $\sin\varphi \,.\, rX$ is divisible by $xx - 2\cos\varphi \,.\, rx + rr$, and hence, unless $r\sin\varphi = 0$, so is the polynomial X. Q. E. P.

II. But if $r\sin\varphi = 0$, either $r = 0$ or $\sin\varphi = 0$. In the first case, $M = 0$ by [1], and thus X is divisible by x, or by $x - r\cos\varphi$; in the second case, $\cos\varphi = \pm 1$, $\cos 2\varphi = +1$, $\cos 3\varphi = \pm 1$, and in general $\cos n\varphi = \cos\varphi^n$. Thus, by [1], we obtain $X = 0$ on setting $x = r\cos\varphi$, and consequently the polynomial X will be divisible by $x - r\cos\varphi$. Q. E. S.

15.

The preceding theorem is usually proved with the aid of imaginary quantities; see EULER, *Introd. in Anal. Inf.*, Vol. I, p. 110. I thought it worth showing how it can be obtained just as easily without their aid. It is now clear that proving our theorem requires nothing more than showing the following: *Given any polynomial X of the form $x^m + Ax^{m-1} + Bx^{m-2} + etc. + Lx + M$, r and φ can be determined so that equations* [1] *and* [2] *hold.* For it follows that X has a real factor of the first or second degree; division will necessarily produce a real quotient of lower degree, which, for the same reason, will also have a factor of the first or second degree. By continuing this operation, X will eventually be resolved into real linear or quadratic factors. The purpose of the following investigations is therefore to prove that theorem.

16.

Imagine a fixed infinite plane (the plane of the plate, Fig. 1), and in it a fixed infinite straight line GC passing through a fixed point C. Take some length as the unit, so that all lengths of straight lines can be expressed by numbers. At any point P of the plane whose distance from the centre C is r and whose angle is $GCP = \varphi$, erect a perpendicular equal to the value of the expression

$$r^m \sin m\varphi + Ar^{m-1}\sin(m-1)\varphi + \text{etc.} + Lr\sin\varphi$$

which, for brevity, I shall always denote by T in what follows. I always regard the distance r as positive; for points lying on the other side of the axis, the angle φ must be regarded either as greater than two right angles or as negative (which here amounts to the same thing). The endpoints of these perpendiculars (which must be taken above the plane for a positive value of T, below it for a negative value, and in the plane itself for a vanishing value) will lie on a continuous curved surface extending infinitely in every direction, which, for brevity, I shall call the *first surface* in what follows. In exactly the same way, refer another surface to the same plane and centre and the same axis, with its height above any point of the plane equal to

$$r^m \cos m\varphi + Ar^{m-1} \cos(m-1)\varphi + \text{etc.} + Lr \cos \varphi + M$$

an expression which, for brevity, I shall always denote by U. I shall distinguish this surface, which will likewise be continuous and extend infinitely in every direction, from the former by calling it the *second surface*. It is then clear that the whole task is to show that there is at least one point lying in the plane, in the first surface and in the second surface at the same time.

17.

It is easy to see that the first surface lies partly above and partly below the plane. For the distance r from the centre can clearly be taken so large that the remaining terms in T vanish in comparison with the first term $r^m \sin m\varphi$; and, by choosing the angle φ appropriately, this term can be made either positive or negative. The fixed plane will therefore necessarily be cut by the first surface. I shall call this intersection of the plane with the first surface the *first line*; it will thus be determined by the equation $T = 0$. For the same reason, the plane will be cut by the second surface; the intersection will form a curve determined by the equation $U = 0$, which I shall call the *second line*. Strictly speaking, each curve will consist of several branches, which may be entirely separate, though each will be a continuous line. Indeed, the first line will always be what is called a compound curve, and the axis GC must be regarded as part of this curve: whatever value is assigned to r, we always have $T = 0$ whenever $\varphi = 0$ or $= 180°$. But it is better to regard the collection of all branches passing through all points where $T = 0$ as a single curve (following the usage generally accepted in higher geometry), and likewise all branches passing through all points where $U = 0$. It is now clear that the task has been reduced to showing that there is at least one point in the plane where some branch of the first line is cut by a branch of the second line. For this purpose, we must examine the nature of these lines more closely.

18.

First of all, I observe that both curves are algebraic and, when referred to orthogonal coordinates, of degree m. For if the origin of abscissas is taken at C, the abscissas x towards G, and the ordinates y towards P, we have $x = r\cos\varphi$, $y = r\sin\varphi$, and thus in general, whatever n may be,

$$r^n \sin n\varphi = nx^{n-1}y - \frac{n\,.\,n-1\,.\,n-2}{1\,.\,2\,.\,3}x^{n-3}y^3 + \frac{n\,\ldots\,n-4}{1\,\ldots\,.5}x^{n-5}y^5 - \text{etc.},$$

$$r^n \cos n\varphi = x^n - \frac{n\,.\,n-1}{1\,.\,2}x^{n-2}yy + \frac{n\,.\,n-1\,.\,n-2\,.\,n-3}{1\,.\,2\,.\,3\,.\,4}x^{n-4}y^4 - \text{etc.}$$

Both T and U will therefore consist of several terms of the form $ax^\alpha y^\beta$, where α and β denote positive integers whose sum, at its greatest, is $= m$. It can also readily be foreseen that all terms of T contain the factor y, and thus that the first line is, strictly speaking, composed of a straight line (whose equation is $y = 0$) and a curve of degree $m - 1$; but we need not take this distinction into account here.

It will be more important to investigate whether the first and second lines have arms extending to infinity, and how many and of what kind. At an infinite distance from the point C, the first line, whose equation is $\sin m\varphi + \frac{A}{r}\sin(m-1)\varphi + \frac{B}{rr}\sin(m-2)\varphi$ etc. $= 0$, will coincide with the line whose equation is $\sin m\varphi = 0$. This represents m straight lines intersecting at the point C, the first of which is the axis GCG', while the others are inclined to it at angles of $\frac{1}{m}180$, $\frac{2}{m}180$, $\frac{3}{m}180$ etc. degrees. The first line therefore has $2m$ infinite branches, which divide the circumference of a circle drawn with infinite radius into $2m$ equal parts: the first branch cuts the circumference at the intersection of the circle and the axis, the second at a distance of $\frac{1}{m}180°$, the third at a distance of $\frac{2}{m}180°$, etc. In the same way, at an infinite distance from the centre the second line will have an asymptote expressed by the equation $\cos m\varphi = 0$. This is a collection of m straight lines likewise intersecting at the point C at equal angles, but in such a way that the first forms an angle of $\frac{1}{m}90°$ with the axis CG, the second an angle of $\frac{3}{m}90°$, the third an angle of $\frac{5}{m}90°$, etc. The second line will therefore also have $2m$ infinite branches, each occupying the middle position between two neighbouring branches of the first line, so that they cut the circumference of a circle drawn with an infinitely large radius at points whose angular distances from the axis are $\frac{1}{m}90°$, $\frac{3}{m}90°$, $\frac{5}{m}90°$, etc. It is also clear that the axis itself always forms two infinite branches of the first line, namely the first and the one in position $m + 1$. This arrangement of the branches is shown very clearly in Fig. 2,

drawn for the case $m = 4$; the branches of the second line are dotted to distinguish them from those of the first line, and the same applies to the fourth figure*). — Since these conclusions are of the greatest importance, however, and infinitely large quantities may trouble some readers, I shall also show in the next article how to obtain them without the aid of infinities.

19.

THEOREM. *With everything as above, a circle can be drawn with centre C, on whose circumference there are $2m$ points where $T = 0$ and as many where $U = 0$, arranged so that each of the latter lies between two of the former.*

Let the sum of all the coefficients A, B etc. K, L, M, taken positively, be $= S$, and take R to be both $> S\sqrt{2}$ and > 1*). I then say that what the theorem states necessarily holds for the circle drawn with radius R. For brevity, label the point of this circumference that is $\frac{1}{m}45$ degrees from its intersection with the left part of the axis, or for which $\varphi = \frac{1}{m}45°$, by (1); similarly, the point that is $\frac{3}{m}45°$ from that intersection, or for which $\varphi = \frac{3}{m}45°$, by (3); then the point where $\varphi = \frac{5}{m}45°$ by (5), and so on up to $(8m-1)$, which is $\frac{8m-1}{m}45$ degrees from that intersection if one always proceeds in the same direction (or $\frac{1}{m}45°$ in the opposite direction). Thus there are altogether $4m$ points on the circumference at equal intervals. Between $(8m-1)$ and (1) there will be one point where $T = 0$; likewise, one such point will lie between (3) and (5), between (7) and (9), between (11) and (13), etc., so that their number is $2m$. In the same way, one point where $U = 0$ will lie between (1) and (3), between (5) and (7), and between (9) and (11), so that their number is also $= 2m$. Finally, apart from these $4m$ points, there will be no others anywhere on the circumference where either T or U is $= 0$.

Proof. I. At the point (1), $m\varphi = 45°$, and thus

$$T = R^{m-1}(R\sqrt{\tfrac{1}{2}} + A\sin(m-1)\varphi + \tfrac{B}{R}\sin(m-2)\varphi + \text{etc.} + \tfrac{L}{R^{m-2}}\sin\varphi)$$

But the sum $A\sin(m-1)\varphi + \frac{B}{R}\sin(m-2)\varphi$ etc. certainly cannot be greater than S, and must therefore be less than $R\sqrt{\frac{1}{2}}$. It follows that at this point the value of T is certainly positive. All the more, then, T will have a positive value when $m\varphi$ lies between 45° and 135°; that is, the value of T will always be positive from the point (1) to (3). For the same reason, T will be positive everywhere from the point (9) to (11), and in general from any point $(8k+1)$ to $(8k+3)$, where k denotes any integer. Similarly, T will be negative everywhere between (5) and (7), between (13) and (15), etc., and in general between

*)The fourth figure was drawn on the assumption $X = x^4 - 2xx + 3x + 10$, so that readers less accustomed to general and abstract investigations can see in a concrete case the relative positions of the two curves. The length of the line CG was taken as $= 10$ ($CN = 1{,}26255$.)

*)When $S > \sqrt{\frac{1}{2}}$, the first condition implies the second; but when $S < \sqrt{\frac{1}{2}}$, the second implies the first.

$(8k+5)$ and $(8k+7)$; thus it cannot be $= 0$ anywhere in any of these intervals. But since its value is positive at (3) and negative at (5), it must be $= 0$ somewhere between (3) and (5); likewise somewhere between (7) and (9), between (11) and (13), etc., up to and including the interval between $(8m-1)$ and (1). Thus we have $T = 0$ at altogether $2m$ points. Q. E. P.

II. That there are no other points with this property apart from these $2m$ points is seen as follows. Since there are none between (1) and (3), between (5) and (7), etc., there could be more such points only if at least two lay in some interval between (3) and (5), or between (7) and (9), etc. Then, however, T would necessarily be a *maximum* or *minimum* somewhere in the same interval, and hence $\frac{\mathrm{d}T}{\mathrm{d}\varphi} = 0$. But $\frac{\mathrm{d}T}{\mathrm{d}\varphi} = mR^{m-2}(R\cos m\varphi + \frac{m-1}{m}A\cos(m-1)\varphi + \text{etc.})$, and between (3) and (5), $\cos m\varphi$ is always negative and $> \sqrt{\frac{1}{2}}$. It is therefore easy to see that $\frac{\mathrm{d}T}{\mathrm{d}\varphi}$ is negative throughout this interval; in the same way, it is positive everywhere between (7) and (9), negative between (11) and (13), etc. Thus it cannot be 0 in any of these intervals, and the supposition cannot stand. Therefore etc. Q. E. S.

III. In exactly the same way, it is proved that U has a negative value everywhere between (3) and (5), between (11) and (13), etc., and in general between $(8k+3)$ and $(8k+5)$; but a positive value between (7) and (9), between (15) and (17), etc., and in general between $(8k+7)$ and $(8k+9)$. It follows at once that $U = 0$ must hold somewhere between (1) and (3), between (5) and (7), etc., that is, at $2m$ points. But $\frac{\mathrm{d}U}{\mathrm{d}\varphi} = 0$ cannot hold in any of these intervals (which is easily proved in the same way as above). There will therefore be no more than those $2m$ points on the circumference of the circle where $U = 0$. Q. E. T. and Q.

The part of the theorem stating that there are no more than $2m$ points where $T = 0$, and no more than $2m$ where $U = 0$, can also be proved from the fact that equations $T = 0$, $U = 0$ represent curves of degree m. For it is known from higher geometry that such curves cannot be cut in more than $2m$ points by a circle, as a curve of the second degree.

20.

If another circle is drawn from the same centre with a radius greater than R and divided in the same way, then on this circle too there will be one point between (3) and (5) where $T = 0$, and likewise between (7) and (9), etc. It is easy to see that the less the radius of this circle differs from the radius R, the nearer such points lying between (3) and (5) on the two circumferences must be to each other. The same will hold if a circle is drawn with a radius somewhat smaller than R, but still greater than $S\sqrt{2}$ and 1. From this it is readily understood that the circumference of the circle drawn with radius R is actually *cut* by some branch of the first line at the point between (3) and (5) where $T = 0$; and the same holds for the remaining points where $T = 0$. In the

same way, it is clear that the circumference of this circle is cut by some branch of the second line at all $2m$ points where $U = 0$. These conclusions may also be expressed as follows: if a circle of the appropriate size is drawn with centre C, $2m$ branches of the first line and as many branches of the second line will enter it, in such a way that each pair of neighbouring branches of the first line is separated by some branch of the second line. See Fig. 2, where the circle will now be of finite rather than infinite size. The numbers attached to the individual branches must not be confused with the numbers by which, for brevity, I designated certain boundaries on the circumference in this and the preceding article.

21.

From this relative arrangement of the branches entering the circle, the necessary existence of an intersection of some branch of the first line with a branch of the second line inside the circle can now be deduced in so many different ways that I almost do not know which method to choose in preference to the others. The following seems the clearest. Let us label (Fig. 2) the point on the circumference where it is cut by the left part of the axis (which is itself one of the $2m$ branches of the first line) by 0; the next point, where a branch of the second line enters, by 1; the point next to this, where the second branch of the first line enters, by 2; and so on up to $4m - 1$. Thus a branch of the second line enters the circle at every point marked with an even number, while a branch of the second line enters at every point marked with an odd number. Now it is known from higher geometry that every algebraic curve (or each individual part of any algebraic curve, if it happens to consist of several parts) either returns into itself or extends to infinity in both directions. Hence, if a branch of an algebraic curve enters a bounded region, it must necessarily leave that region again somewhere[*)]. From this it is easily concluded that every point marked with an even number (or, for

[*)]It certainly seems to have been demonstrated well enough that an algebraic curve can neither suddenly break off anywhere (as happens, for example, with the transcendental curve whose equation is $y = \frac{1}{\log x}$), nor, after infinitely many windings, as it were lose itself at some point (like the logarithmic spiral); and, as far as I know, no one has raised a doubt on this matter. Nevertheless, if anyone demands it, I shall undertake to give a proof free from all doubt on another occasion. In the present case, however, it is clear that if some branch, for example 2, never left the circle anywhere (Fig. 3), you could enter the circle between 0 and 2, then go around the whole of this branch (which would have to lose itself in the region of the circle), and finally leave the circle again between 2 and 4, without encountering the first line anywhere along the entire path. That this is absurd is clear from the fact that, at the point where you entered the circle, the first surface was above you, and at your exit it was below you. You must therefore necessarily have encountered the first surface itself somewhere, or a point of the first line. — This reasoning, which rests on principles of the geometry of position that are no less valid than those of the geometry of magnitude, implies only that, if you enter the circle along some branch of the first line, you can leave it again at another place while always remaining on the first line; it does not imply that your path is a continuous line in the

brevity, *every even point*) must be joined within the circle to another even point by a branch of the first line; similarly, every point marked with an odd number must be joined to another such point by a branch of the second line. Although this connection of pairs of points may vary greatly with the nature of the polynomial X, so that it cannot be determined in general, it is nevertheless easy to prove that *whatever it may be, an intersection of the first line with the second line always results.*

22.

It seems that this necessity can most conveniently be proved by contradiction. Suppose, then, that each pair of even points and each pair of odd points can be joined in such a way that no intersection of a branch of the first line with a branch of the second line results. Since the axis is part of the first line, the point 0 will clearly be joined to the point $2m$. The point 1 therefore cannot be joined to any point lying beyond the axis, that is, to any point designated by a number greater than $2m$; for otherwise the joining line would necessarily cut the axis. Thus, if 1 is supposed to be joined to the point n, we shall have $n < 2m$. For a similar reason, if 2 is taken to be joined to n', we shall have $n' < n$; for otherwise the branch $2 \ldots n'$ would necessarily cut the branch $1 \ldots n$. For the same reason, the point 3 will be joined to one of the points lying between 4 and n'. It is clear that if 3, 4, 5, etc. are supposed to be joined to n'', n''', n'''', etc., then n''' lies between 5 and n'', n'''' between 6 and n''', etc. It is therefore evident that we shall eventually reach some point h joined to the point $h + 2$. Then the branch entering the circle at the point $h + 1$ will necessarily cut the branch joining the points h and $h + 2$. Since one of these two branches belongs to the first line and the other to the second, it is now clear that the supposition is contradictory, and hence that an intersection of the first line with the second line must occur somewhere. Combining this with what precedes, we conclude from all the investigations set out above that the theorem *that every polynomial in one variable can be resolved into real factors of the first or second degree* has been proved with full rigour.

23.

It is also not difficult to deduce from the same principles that there is not merely one intersection of the first line with the second, but at least m. It may, however, happen that the first line is cut by several branches of the second line at the same point, in which case the polynomial X will have several equal factors. But since it is enough here to have proved the necessity of one intersection, I shall not dwell on this further, for brevity. For the same reason, I shall not pursue other properties of these lines in more detail here:

sense accepted in higher geometry. But here it is enough for the path to be a continuous line in the ordinary sense, that is, nowhere interrupted but connected everywhere.

for example, that their intersections always occur at right angles; or that, if several arms of both curves meet at the same point, there will be as many arms of the first line as of the second, arranged alternately and intersecting at equal angles, etc.

Finally, I observe that it is by no means impossible to present the preceding proof, which I have here built on geometric principles, in a purely analytic form as well. But I thought that the presentation set out here would be less abstract, and that the true force of the proof is brought much more clearly before the eyes here than could be expected from an analytic proof.

To finish, I shall also indicate another method of proving our theorem. At first sight it will seem very different, not only from the preceding proof, but also from all the other proofs described above; yet, if one looks at its essence, it is in fact the same as D'ALEMBERT's. I leave it to the experts to compare it with his and to examine the parallel between the two; it is added solely for their sake.

24.

Above the plane of Fig. 4, I suppose the first and second surfaces to be drawn in the same way as above, relative to the axis CG and the fixed point C. Take any point on some branch of the first line, that is, where $T = 0$ (for example, any point M lying on the axis). Unless $U = 0$ holds there too, proceed from this point along the first line in the direction in which the absolute value of U decreases. If at the point M the absolute value of U happens to decrease in both directions, it is immaterial which way you proceed; I shall show at once what is to be done if U increases in both directions. It is therefore clear that, as you continue along the first line, you must eventually reach either a point where $U = 0$, or one where the value of U becomes a minimum, for example the point N. In the former case, what was sought has been found. In the latter, it can be proved that several branches of the first line intersect at this point (an even number of branches, in fact), half of which are such that, if you turn into one of them (in either direction), the value of U continues to decrease. (For brevity, I must omit the proof of this theorem, which is more lengthy than difficult.) You can therefore proceed again along this branch until U either becomes $= 0$ (as happens at P in Fig. 4) or again becomes a minimum. Then you will turn aside again, and must eventually reach a point where $U = 0$.

One might raise against this proof the doubt whether it is possible that, however far you proceed and although the value of U always decreases, these decreases nevertheless become continually slower, and that value never reaches some limit. This objection would correspond to the fourth in Article 6. But it would not be difficult to assign a boundary beyond which the value of U must not only change ever more rapidly, but can no longer *decrease*, so that the value 0 must necessarily already have occurred before you reach this boundary. I reserve for another occasion a fuller treatment of this and the remaining matters that I have only been able to indicate in this proof.

We discovered the principles on which this proof rests at the beginning of October 1797.

Fig 1.

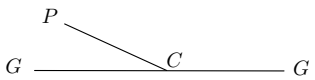


Fig.2.

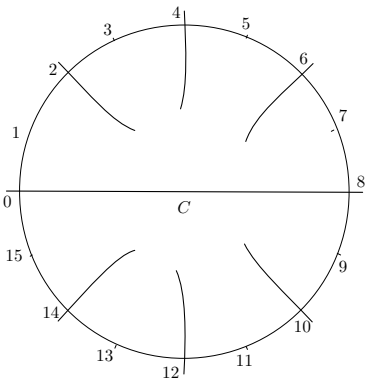


Fig 4

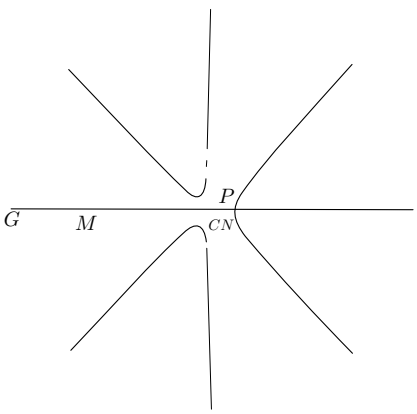
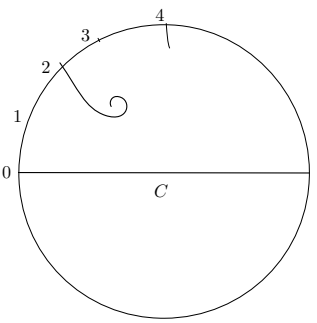


Fig 3



ANOTHER NEW PROOF

OF THE THEOREM

THAT EVERY POLYNOMIAL

IN ONE VARIABLE

CAN BE RESOLVED INTO REAL FACTORS

OF THE FIRST OR SECOND DEGREE

B Y

CARL FRIEDRICH GAUSS

SUBMITTED TO THE ROYAL SOCIETY OF SCIENCES, 1815,
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ANOTHER NEW PROOF OF THE THEOREM

THAT EVERY POLYNOMIAL

IN ONE VARIABLE

CAN BE RESOLVED INTO REAL FACTORS OF THE FIRST OR SECOND DEGREE.

1.

Although the proof of the theorem on resolving polynomials into factors that I gave in a paper published sixteen years ago seems to leave nothing to be desired either in rigour or in simplicity, I hope that mathematicians will not find it unwelcome if I return to this most important question and attempt to establish another proof, no less rigorous, from entirely different principles. That earlier proof depends, at least in part, on geometric considerations; the one I now undertake to set out, by contrast, will rest on purely analytic principles. In the work cited, I reviewed the more notable analytic methods by which other mathematicians had tried to prove our theorem up to that time, and gave a detailed account of the defects they suffer from. The most serious and truly fundamental of these is common to all those attempts, as well as to the more recent ones that have come to my attention; yet even then I declared that it seemed by no means unavoidable in an analytic proof. Let the experts now judge whether these further efforts have fully redeemed the pledge I gave at that time.

2.

Some preliminary investigations will precede the main discussion, both so that nothing may seem to be lacking and because their treatment may perhaps shed some fresh light even on matters that others have already touched upon. We shall first consider the greatest common divisor of two polynomials in one indeterminate. It should be stated at the outset that we are always speaking here only of polynomials: if a product is formed from two of them, each is called a *divisor* of that product. The *degree* of a divisor is determined by the exponent of the highest power of the indeterminate it contains, without any regard at all to the numerical coefficients. We can deal with the remaining matters concerning common divisors of polynomials more briefly, since they are entirely analogous to the corresponding properties of common divisors of numbers.

Given two polynomials Y, Y' in the indeterminate x, of which the former has a higher degree than the latter, or at least not a lower one, we shall form the following equations:

$$\begin{gathered} Y = qY' + Y'' \\ Y' = q'Y'' + Y''' \\ Y'' = q''Y''' + Y'''' \\ \text{etc., up to} \\ Y^{(\mu-1)} = q^{(\mu-1)}Y^{(\mu)} \end{gathered}$$

The rule is to divide Y first by Y' in the usual way; then Y' by the remainder Y'' of the first division, which has lower degree than Y'; then again the first remainder by the second, Y'''; and so on until a division without remainder is reached. That this must eventually happen follows from the continually decreasing degrees of the polynomials Y', Y'', Y''' etc. There is hardly any need to point out that these functions, as well as the quotients q, q', q'' etc., are *polynomials* in x. With this established, it is clear that:

I. Going backwards from the last of these equations to the first, the polynomial $Y^{(\mu)}$ is seen to divide each of the preceding ones, and is therefore certainly a common divisor of the given polynomials Y, Y'.

II. Going forwards from the first equation to the last, we see that every common divisor of the polynomials Y, Y' also divides each of the following ones, and hence the last, $Y^{(\mu)}$. Thus the polynomials Y, Y' cannot have any common divisor of higher degree than $Y^{(\mu)}$. Every common divisor of the same degree as $Y^{(\mu)}$ will stand to it in the ratio of a number to a number; hence it must itself be taken as the greatest common divisor.

III. If $Y^{(\mu)}$ has degree 0, that is, is a number, no function of the indeterminate x in the proper sense can divide Y, Y'. In this case, therefore, these polynomials are said to have no common divisor.

IV. Take the equation immediately before the last one; then eliminate $Y^{(\mu-1)}$ from it with the aid of the equation immediately before that; then again eliminate $Y^{(\mu-2)}$ with the aid of the preceding equation, and so on. In this way we obtain

$$\begin{aligned} Y^{(\mu)} &= +kY^{(\mu-2)} - k'Y^{(\mu-1)} \\ &= -k'Y^{(\mu-3)} + k''Y^{(\mu-2)} \\ &= +k''Y^{(\mu-4)} - k'''Y^{(\mu-3)} \\ &= -k'''Y^{(\mu-5)} + k''''Y^{(\mu-4)} \\ &\quad \text{etc.} \end{aligned}$$

if we suppose the polynomials k, k', k'' etc. to be formed by the following rule:

$$\begin{aligned} k &= 1 \\ k' &= q^{(\mu-2)} \\ k'' &= q^{(\mu-3)}k' + k \\ k''' &= q^{(\mu-4)}k'' + k' \\ k'''' &= q^{(\mu-5)}k''' + k'' \\ &\quad \text{etc.} \end{aligned}$$

We shall therefore have

$$\pm k^{(\mu-2)}Y \mp k^{(\mu-1)}Y' = Y^{(\mu)}$$

with the upper signs applying when μ is even and the lower when it is odd. Thus, when Y and Y' have no common divisor, we can in this way find two polynomials Z, Z' in the indeterminate x such that

$$ZY + Z'Y' = 1$$

V. The converse of this proposition clearly holds as well: if the equation

$$ZY + Z'Y' = 1$$

can be satisfied with Z, Z' polynomials in the indeterminate x, then Y and Y' certainly cannot have a common divisor.

3.

The second preliminary investigation will concern the transformation of symmetric functions. Let a, b, c etc. be indeterminate quantities, m in number. Denote their sum by λ', the sum of their products in pairs by λ'', the sum of their products in threes by λ''', etc., so that expanding the product

$$(x-a)(x-b)(x-c)\dots.$$

gives

$$x^m - \lambda' x^{m-1} + \lambda'' x^{m-2} - \lambda''' x^{m-3} + \text{etc.}$$

Thus $\lambda', \lambda'', \lambda'''$ etc. are symmetric functions of the indeterminates a, b, c etc.: that is, functions in which these indeterminates occur in the same way, or more explicitly, functions unchanged by any permutation of these indeterminates among themselves. More generally, every polynomial in $\lambda', \lambda'', \lambda'''$ etc. (whether it contains only these indeterminates or also others independent of a, b, c etc.) will clearly be a symmetric polynomial in the indeterminates a, b, c etc.

4.

The converse theorem is somewhat less evident. Let ρ be a symmetric function of the indeterminates a, b, c etc.; it will thus consist of a certain number of terms of the form

$$Ma^\alpha b^\beta c^\gamma \dots.$$

where α, β, γ etc. denote nonnegative integers, and M is a coefficient either fixed or at least independent of a, b, c etc. (if the function ρ happens to contain other indeterminates besides a, b, c etc.). First we shall establish a definite ordering of these individual terms. For this purpose, we first arrange the indeterminates a, b, c etc. in a definite order, which is itself entirely arbitrary: for example, with a in the first place, b in the second, c in the third, etc. Then, of the two terms

$$Ma^\alpha b^\beta c^\gamma \dots. \quad \text{and} \quad Ma^{\alpha'} b^{\beta'} c^{\gamma'} \dots.$$

we shall place the former higher in the ordering than the latter if

$$\begin{gathered} \text{or} \quad \alpha > \alpha', \quad \text{or} \quad \alpha = \alpha' \text{ and } \beta > \beta', \\ \text{or} \quad \alpha = \alpha', \ \beta = \beta' \text{ and } \gamma > \gamma', \quad \text{or etc.} \end{gathered}$$

that is, if the first nonvanishing one among the differences $\alpha - \alpha', \beta - \beta', \gamma - \gamma'$ etc. is positive. Terms in the same place in this ordering therefore differ only in their coefficient M, and can be combined into one term. We

shall accordingly suppose the individual terms of the function ρ to occupy different places in the ordering.

Now observe that, if $Ma^\alpha b^\beta c^\gamma \ldots\ldots$ is the highest of all the terms of the function ρ in this ordering, α must be greater than, or at least not less than, β. For if $\beta > \alpha$, the term $Ma^\beta b^\alpha c^\gamma \ldots\ldots$, which the function ρ must also contain since it is symmetric, would stand higher than $Ma^\alpha b^\beta c^\gamma \ldots\ldots$, contrary to the hypothesis. In the same way, β will be greater than, or at least not less than, γ; further, γ will not be less than the next exponent δ, etc. Hence each of the differences $\alpha - \beta$, $\beta - \gamma, \gamma - \delta$ etc. is a nonnegative integer.

Secondly, consider a product formed from any number of polynomials in the indeterminates a, b, c etc. Its highest term must be the product of the highest terms of those factors. It is equally clear that the highest terms of the functions $\lambda', \lambda'', \lambda'''$ etc. are respectively a, ab, abc etc. It follows that the highest term arising from the product

$$p = M\lambda'^{\alpha-\beta}\lambda''^{\beta-\gamma}\lambda'''^{\gamma-\delta}\ldots\ldots$$

is $Ma^\alpha b^\beta c^\gamma \ldots\ldots$. Thus, on setting $\rho - p = \rho'$, the highest term of the function ρ' will certainly stand lower in the ordering than the highest term of the function ρ. But p, and therefore also ρ', are clearly symmetric polynomials in a, b, c etc. Treating ρ' in the same way as ρ before it, we can therefore split it into $p' + \rho''$, where p' is a product of powers of $\lambda', \lambda'', \lambda'''$ etc. multiplied by a coefficient either fixed or at least independent of a, b, c etc., while ρ'' is a symmetric polynomial in a, b, c etc. whose highest term stands lower in the ordering than the highest term of ρ'. Continuing in the same way, ρ will clearly at last be reduced to the form $p + p' + p'' + p'''$ etc., that is, transformed into a polynomial in $\lambda', \lambda'', \lambda'''$ etc.

5.

We can also state the theorem proved in the preceding article as follows. Given any symmetric polynomial ρ in the indeterminates a, b, c etc., we can specify a polynomial in as many other indeterminates l', l'', l''' etc. which becomes ρ under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc. It is also easy to show that *this can be done in only one way.* For suppose that two different functions of the indeterminates l', l'', l''' etc., namely r and r', yield the same function of a, b, c etc. after the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc. Then $r - r'$ will be a function of l', l'', l''' etc. which does not vanish in itself, but vanishes identically after those substitutions. We shall readily see that this is absurd if we consider that $r - r'$ must consist of a certain number of parts of the form

$$Ml'^\alpha l''^\beta l'''^\gamma \ldots\ldots$$

whose coefficients M do not vanish and which all differ from one another in their exponents.

The highest terms arising from these individual parts are therefore given by

$$Ma^{\alpha+\beta+\gamma+\text{etc.}}b^{\beta+\gamma+\text{etc.}}c^{\gamma+\text{etc.}}\ldots\ldots$$

and must consequently occupy different places in the ordering, so that the highest term of all cannot possibly cancel.

The actual calculation for transformations of this kind could, moreover, be greatly shortened by various shortcuts. We do not dwell on them here, since the mere possibility of the transformation already suffices for our purpose.

6.

Consider the product of $m(m-1)$ factors

$$\begin{aligned}&(a-b)(a-c)(a-d)\ldots\ldots\\ \times&(b-a)(b-c)(b-d)\ldots\ldots\\ \times&(c-a)(c-b)(c-d)\ldots\ldots\\ \times&(d-a)(d-b)(d-c)\ldots\ldots\\ &\text{etc.}\end{aligned}$$

which we shall denote by π. Since it involves the indeterminates a, b, c etc. symmetrically, we shall suppose it reduced to the form of a polynomial in $\lambda', \lambda'', \lambda'''$ etc. Let this polynomial become p when l', l'', l''' etc. are substituted respectively for $\lambda', \lambda'', \lambda'''$ etc. With this established, we shall call p the *determinant* of the polynomial

$$y = x^m - l'x^{m-1} + l''x^{m-2} - l'''x^{m-3} + \text{etc.}$$

Thus, for example, for $m = 2$ we have

$$p = -l'^2 + 4l''$$

Likewise, for $m = 3$ we find

$$p = -l'^2l''^2 + 4l'^3l''' + 4l''^3 - 18l'l''l''' + 27l'''^2$$

The determinant of the polynomial y is therefore a polynomial in the coefficients l', l'', l''' etc. which, under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc., becomes the product of all the differences between pairs of the quantities a, b, c etc. In the case $m = 1$, that is, when there is only one indeterminate a and hence no differences at all, it will be convenient to take the number 1 itself as the determinant of the polynomial y.

In establishing the notion of the determinant, we had to regard the coefficients of the polynomial y as indeterminate quantities. The determinant of a polynomial with fixed coefficients,

$$Y = x^m - L'x^{m-1} + L''x^{m-2} - L'''x^{m-3} + \text{etc.}$$

will be a fixed number P, namely the value of the polynomial p for $l' = L', l'' = L''$, $l''' = L'''$ etc. Thus, if we suppose that Y can be resolved into linear factors,

$$Y = (x - A)(x - B)(x - C)\ldots.$$

or that Y arises from

$$\upsilon = (x - a)(x - b)(x - c)\ldots.$$

by setting $a = A, b = B, c = C$ etc., so that the same substitutions turn $\lambda', \lambda'', \lambda'''$ etc. respectively into L', L'', L''' etc., then P will clearly equal the product of the factors

$$\begin{aligned}
&(A - B)(A - C)(A - D)\ldots. \\
\times&(B - A)(B - C)(B - D)\ldots. \\
\times&(C - A)(C - B)(C - D)\ldots. \\
\times&(D - A)(D - B)(D - C)\ldots. \\
&\text{etc.}
\end{aligned}$$

It follows that, if $P = 0$, at least two of the quantities A, B, C etc. must be equal; conversely, if it is not the case that $P = 0$, all the quantities A, B, C etc. must be unequal. We now observe that, on setting $\frac{\mathrm{d}Y}{\mathrm{d}x} = Y'$, or

$$\begin{aligned}
Y' = mx^{m-1} - (m-1)L'x^{m-2} + (m-2)L''x^{m-3} \\
-(m-3)L'''x^{m-4} + \text{etc.}
\end{aligned}$$

we have

$$\begin{aligned}
Y' = &(x - B)(x - C)(x - D)\ldots. \\
&+ (x - A)(x - C)(x - D)\ldots. \\
&+ (x - A)(x - B)(x - D)\ldots. \\
&+ (x - A)(x - B)(x - C)\ldots. \\
&+ \text{etc.}
\end{aligned}$$

Thus, if two of the quantities A, B, C etc. are equal, for example $A = B$, then Y' will be divisible by $x - A$; in other words, Y and Y' will have the common divisor $x - A$. Conversely, if Y' is supposed to have any common divisor with Y, then Y' must contain one of the linear factors $x-A, x-B, x-C$ etc., for example the first, $x - A$. This clearly cannot happen unless A equals one of the remaining quantities B, C, D etc. From all this we therefore obtain two THEOREMS:

I. *If the determinant of the polynomial Y is $= 0$, then Y certainly has a common divisor with Y'; hence, if Y and Y' have no common divisor, the determinant of the polynomial Y cannot be $= 0$.*

II. *If the determinant of the polynomial Y is not $= 0$, then Y and Y' certainly cannot have a common divisor; or, if Y and Y' have a common divisor, the determinant of the polynomial Y must necessarily be $= 0$.*

7.

But it must be carefully noted that the whole force of this very simple proof rests on the supposition that the polynomial Y can be resolved into linear factors. At this very point, where the general possibility of such a resolution is to be proved, that supposition would be nothing but begging the question. Yet not all those who have attempted analytic proofs of the main theorem have guarded against fallacies exactly like this. We have already pointed out the origin of this plausible illusion in the very statement of the inquiry: all of them investigated only the *form* of the roots of equations, whereas they ought to have proved their *existence*, which they rashly assumed. But enough has already been said in the paper cited above about this way of proceeding, which departs too far from rigour and clarity. We shall therefore now place the theorems of the preceding article, at least one of which we cannot do without for our purpose, on a firmer foundation. We shall begin with the second, as the easier one.

8.

Denote by ρ the function

$$\begin{aligned}
&\frac{\pi(x-b)(x-c)(x-d)\ldots.}{(a-b)^2(a-c)^2(a-d)^2\ldots.}\\
&+\frac{\pi(x-a)(x-c)(x-d)\ldots.}{(b-a)^2(b-c)^2(b-d)^2\ldots.}\\
&+\frac{\pi(x-a)(x-b)(x-d)\ldots.}{(c-a)^2(c-b)^2(c-d)^2\ldots.}\\
&+\frac{\pi(x-a)(x-b)(x-c)\ldots.}{(d-a)^2(d-b)^2(d-c)^2\ldots.}\\
&+\text{etc.}
\end{aligned}$$

which, since π is divisible by each denominator, is a polynomial in the indeterminates x, a, b, c etc. Further, set $\frac{\mathrm{d}\upsilon}{\mathrm{d}x} = \upsilon'$, so that we have

$$\begin{aligned}
\upsilon' = &(x-b)(x-c)(x-d)\ldots.\\
&+(x-a)(x-c)(x-d)\ldots.\\
&+(x-a)(x-b)(x-d)\ldots.\\
&+(x-a)(x-b)(x-c)\ldots.\\
&+\text{etc.}
\end{aligned}$$

For $x = a$ we clearly have $\rho\upsilon' = \pi$. Hence we conclude that the polynomial $\pi - \rho\upsilon'$ is identically divisible by $x - a$, and likewise by $x - b, x - c$ etc., as well as by their product υ. Thus, setting

$$\frac{\pi - \rho\upsilon'}{\upsilon} = \sigma$$

we obtain for σ a polynomial in the indeterminates x, a, b, c etc. which, like ρ, is symmetric with respect to the indeterminates a, b, c etc. We can therefore find two polynomials r, s in the indeterminates x, l', l'', l''' etc. which, under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc., become ρ, σ respectively. Following the analogy, let us denote the polynomial

$$\begin{aligned}
mx^{m-1} - (m-1)l'x^{m-2} + (m-2)l''x^{m-3}\\
-(m-3)l'''x^{m-4} + \text{etc.}
\end{aligned}$$

that is, the derivative $\frac{\mathrm{d}y}{\mathrm{d}x}$, by y', so that y' becomes υ' under those same substitutions. Then $p - sy - ry'$ clearly becomes $\pi - \sigma\upsilon - \rho\upsilon'$ under the same substitutions, that is, 0, and must therefore already vanish identically in itself (Article 5). We thus have the identity

$$p = sy + ry'$$

Hence, if we suppose that substituting $l' = L', l'' = L'', l''' = L'''$ etc. gives $r = R, s = S$, we also have identically

$$P = SY + RY'$$

Here S, R are polynomials in x, while P is a fixed quantity or number. It follows at once that Y and Y' cannot have a common divisor unless $P = 0$. This is precisely the second theorem of Article 6.

9.

We shall complete the proof of the first theorem by showing that, when Y and Y' have no common divisor, $P = 0$ certainly cannot hold. For this purpose, first suppose that two polynomials in the indeterminate x, namely fx and φx, have been found by the rules of Article 2, such that the identity

$$fx \,.\, Y + \varphi x \,.\, Y' = 1$$

holds. We write it here as

$$fx \,.\, \upsilon + \varphi x \,.\, \upsilon' = 1 + fx \,.\, (\upsilon - Y) + \varphi x \,.\, \frac{\mathrm{d}(\upsilon - Y)}{\mathrm{d}x}$$

or, since we have

$$\begin{aligned} \upsilon' = {} & (x-b)(x-c)(x-d)\ldots. \\ & + (x-a) \,.\, \frac{\mathrm{d}[(x-b)(x-c)(x-d)\ldots.]}{\mathrm{d}x} \end{aligned}$$

in the following form:

$$\begin{aligned} & \varphi x \,.\, (x-b)(x-c)(x-d)\ldots. \\ & + \varphi x \,.\, (x-a) \,.\, \frac{\mathrm{d}[(x-b)(x-c)(x-d)\ldots.]}{\mathrm{d}x} \\ & + fx \,.\, (x-a)(x-b)(x-c)(x-d)\ldots. \\ & = 1 + fx \,.\, (\upsilon - Y) + \varphi x \,.\, \frac{\mathrm{d}(\upsilon - Y)}{\mathrm{d}x} \end{aligned}$$

For brevity, we shall denote

$$fx \,.\, (y - Y) + \varphi x \,.\, \frac{\mathrm{d}(y - Y)}{\mathrm{d}x}$$

which is a polynomial in the indeterminates x, l', l'', l''' etc.,

$$\text{by} \quad F(x, l', l'', l''' \text{ etc.})$$

Thus we have identically

$$1 + fx\,.\,(\upsilon - Y) + \varphi x\,.\,\frac{\mathrm{d}(\upsilon - Y)}{\mathrm{d}x}$$
$$= 1 + F(x, \lambda', \lambda'', \lambda''' \text{ etc.})$$

We shall therefore have the identities [1]

$$\varphi a\,.\,(a-b)(a-c)(a-d)\ldots.$$
$$= 1 + F(a, \lambda', \lambda'', \lambda''' \text{ etc.})$$
$$\varphi b\,.\,(b-a)(b-c)(b-d)\ldots.$$
$$= 1 + F(b, \lambda', \lambda'', \lambda''' \text{ etc.})$$
$$\varphi c\,.\,(c-a)(c-b)(c-d)\ldots.$$
$$= 1 + F(c, \lambda', \lambda'', \lambda''' \text{ etc.})$$
$$\text{etc.}$$

Now suppose the product of all the expressions

$$1 + F(a, l', l'', l''' \text{ etc.})$$
$$1 + F(b, l', l'', l''' \text{ etc.})$$
$$1 + F(c, l', l'', l''' \text{ etc.})$$
$$\text{etc.}$$

which is a polynomial in the indeterminates a, b, c etc., l', l'', l''' etc., and is symmetric with respect to a, b, c etc., to be represented by

$$\psi(\lambda', \lambda'', \lambda''' \text{ etc.},\ l', l'', l''' \text{ etc.})$$

Multiplying all the equations [1] then gives the new identity [2]

$$\pi\varphi a\,.\,\varphi b\,.\,\varphi c\ldots. = \psi(\lambda', \lambda'', \lambda''' \text{ etc.},\ \lambda', \lambda'', \lambda''' \text{ etc.})$$

Further, since the product $\varphi a\,.\,\varphi b\,.\,\varphi c\ldots.$ involves the indeterminates a, b, c etc. symmetrically, we can clearly find a polynomial in the indeterminates l', l'', l''' etc. which, under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc., becomes $\varphi a\,.\,\varphi b\,.\,\varphi c\ldots..$ Let t be this polynomial. We then also have identically [3]

$$pt = \psi(l', l'', l''' \text{ etc.},\ l', l'', l''')$$

since these same substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc. turn this equation into the identity [2].

Now the very definition of the function F gives the identity

$$F(x, L', L'', L''' \text{ etc.}) = 0$$

Hence we also have identically

$$1 + F(a, L', L'', L''' \text{ etc.}) = 1$$
$$1 + F(b, L', L'', L''' \text{ etc.}) = 1$$
$$1 + F(c, L', L'', L''' \text{ etc.}) = 1$$
$$\text{etc.}$$

and therefore also identically

$$\psi(\lambda', \lambda'', \lambda''' \text{ etc.},\ L', L'', L''' \text{ etc.}) = 1$$

and thus identically [4]

$$\psi(l', l'', l''' \text{ etc.},\ L'. L'', L''' \text{ etc.}) = 1$$

Combining equations [3] and [4], and substituting $l' = L'$, $l'' = L'', l''' = L'''$ etc., we shall therefore have [5]

$$PT = 1$$

where T denotes the value of the polynomial t corresponding to those substitutions. Since this value must be a finite quantity, P certainly cannot be $= 0$. Q. E. D.

10.

It is now clear from the preceding results that any polynomial Y in one indeterminate x whose determinant is $= 0$ can be decomposed into factors none of which has determinant 0. For, once the greatest common divisor of the polynomials Y and $\frac{\mathrm{d}Y}{\mathrm{d}x}$ has been found, the former is resolved into two factors. If one of these factors[*)] again has determinant 0, it will be resolved into two factors in the same way. We shall continue in this manner until Y is finally resolved into factors none of which has determinant 0.

It is also easy to see that among these factors into which Y is resolved, at least one must have the following property: among the factors of the number expressing its degree, the number two occurs no more often than among the factors of the number m expressing the degree of the polynomial Y. That is, if we set $m = k \,.\, 2^{\mu}$, with k denoting an odd number, then among the factors of the polynomial Y there is at least one of degree $k' \,.\, 2^{\nu}$, where k' is also odd, and either $\nu = \mu$ or $\nu < \mu$. This assertion follows at once from the fact that m is the sum of the numbers expressing the degrees of the individual factors of Y.

[*)] In fact, only the factor which is that common divisor can have determinant 0. But a proof of this proposition would lead us into some detours here; nor is it needed here, since the other factor, even if its determinant could also vanish, could be treated in the same way and resolved into factors.

11.

Before proceeding further, we shall explain an expression whose introduction is of great use in all investigations of symmetric functions, and which will also be very convenient for us. Suppose M is a function of some of the indeterminates a, b, c etc., and let μ be the number of them entering the expression M, without regard to any other indeterminates that M may also involve. Permuting those μ indeterminates in every possible way, both among themselves and with the remaining $m - \mu$ of a, b, c etc., produces from M other expressions similar to M. In all there are

$$m(m-1)(m-2)(m-3)\ldots.(m-\mu+1)$$

expressions, including M itself; we shall call their collection simply the *collection of all M.* It is then clear what is meant by the sum of all M, the product of all M, etc. Thus, for example, π is called the product of all $a - b$, υ the product of all $x - a$, and υ' the sum of all $\frac{\upsilon}{x-a}$, etc.

If M is symmetric with respect to some of the μ indeterminates it contains, permutations of these among themselves do not change the function M. Thus every term occurs several times in the collection of all M: namely, $1\,.\,2\,.\,3\ldots.\nu$ times, if ν is the number of indeterminates with respect to which M is symmetric. If M is symmetric not only with respect to ν indeterminates, but also with respect to ν' others, and again with respect to ν'' others, etc., then M remains unchanged whether two of the first ν indeterminates are interchanged, or two of the second ν', or two of the third ν'', etc. Thus

$$1\,.\,2\,.\,3\ldots.\nu\,.\quad 1\,.\,2\,.\,3\ldots.\nu'\,.\quad 1\,.\,2\,.\,3\ldots.\nu''\text{ etc.}$$

permutations always give identical terms. If we retain only one of these identical terms each time, we therefore obtain in all

$$\frac{m(m-1)(m-2)(m-3)\ldots.(m-\mu+1)}{1\,.\,2\,.\,3\ldots.\nu\,.\,1\,.\,2\,.\,3\ldots.\nu'\,.\,1\,.\,2\,.\,3\ldots.\nu''\text{ etc.}}$$

terms, whose collection we shall call the *collection of all M with repetitions excluded*, to distinguish it from the *collection of all M with repetitions included*. Unless an explicit qualification is given, we shall always understand repetitions to be included.

It is also easy to see that the sum of all M, or the product of all M, or more generally any symmetric function of all M, is always a symmetric function of the indeterminates a, b, c etc., whether repetitions are included or excluded.

12.

With u, x denoting indeterminates, consider now the product of all $u - (a+b)x + ab$, with repetitions excluded, and denote it by ζ. Thus ζ is the product of the following $\frac{1}{2}m(m-1)$ factors:

$$\begin{gathered} u-(a+b)x+ab \\ u-(a+c)x+ac \\ u-(a+d)x+ad \\ \text{etc.} \\ u-(b+c)x+bc \\ u-(b+d)x+bd \\ \text{etc.} \\ u-(c+d)x+cd \\ \text{etc. etc.} \end{gathered}$$

Since this function involves the indeterminates a, b, c etc. symmetrically, we can find a polynomial in the indeterminates u, x, l', l'', l''' etc., to be denoted by z, which becomes ζ when $\lambda', \lambda'', \lambda'''$ etc. are substituted for the indeterminates l', l'', l''' etc. Finally, denote by Z the function of the indeterminates u, x alone into which z passes when the indeterminates l', l'', l''' etc. are assigned the fixed values L', L'', L''' etc.

These three functions ζ, z, Z can be regarded as polynomials of degree $\frac{1}{2}m(m-1)$ in the indeterminate u with indeterminate coefficients. These coefficients are

for ζ, functions of the indeterminates x, a, b, c etc.
for z, functions of the indeterminates x, l', l'', l''' etc.
for Z, functions only of the indeterminate x.

The individual coefficients of z become those of ζ under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc., and those of Z under the substitutions $l' = L', l'' = L'', l''' = L'''$ etc. What we have just said about the coefficients also holds for the determinants of the functions ζ, z, Z. We shall now examine these more closely, with the aim of proving the following

THEOREM. *Whenever $P = 0$ does not hold, the determinant of the function Z certainly cannot be identically $= 0$.*

13.

The proof of this theorem would indeed be very easy if we were allowed to suppose that Y could be resolved into linear factors

$$(x-A)(x-B)(x-C)(x-D)\ldots.$$

For it would then also be certain that Z is the product of all $u-(A+B)x+AB$, and that the determinant of the function Z is the product of the differences between pairs of the quantities

$$\begin{gathered}(A+B)x-AB\\(A+C)x-AC\\(A+D)x-AD\\\text{etc.}\\(B+C)x-BC\\(B+D)x-BD\\\text{etc.}\\(C+D)x-CD\\\text{etc. etc}\end{gathered}$$

But this product cannot vanish identically unless one of its factors itself becomes identically $=0$. This would imply that two of the quantities A, B, C etc. are equal, and thus that the determinant P of the polynomial Y becomes $=0$, contrary to the hypothesis.

Setting aside this argument, which clearly starts by begging the question just as in Article 6, we now proceed directly to give a sound proof of the theorem of Article 12.

14.

The determinant of the function ζ will be the product of all the differences between pairs of $(a+b)x-ab$. The number of these differences is

$$\frac{1}{2}m(m-1)(\tfrac{1}{2}m(m-1)-1)=\frac{1}{4}(m+1)m(m-1)(m-2)$$

This number therefore gives the degree of the determinant of the function ζ with respect to the indeterminate x. The determinant of the function z will have the same degree. The determinant of the function Z, however, may have a lower degree, whenever some coefficients, starting from the highest power of x, vanish. Our task is now to prove that *all* the coefficients in the determinant of the function Z certainly cannot vanish.

On examining more closely the differences whose product is the determinant of the function ζ, we find that some of them—the differences between pairs of $(a+b)x-ab$ having an element in common—give

$$\text{the product of all}\quad (a-b)(x-c)$$

while the remaining ones—the differences between pairs of $(a+b)x-ab$ whose elements are distinct—give

the product of all $\quad (a+b-c-d)x-ab+cd, \quad$ with repetitions excluded.

The first product clearly contains each factor $a-b$ exactly $m-2$ times, and each factor $x-c$ exactly $(m-1)(m-2)$ times. We therefore readily conclude that this product is

$$= \pi^{m-2} \upsilon^{(m-1)(m-2)}$$

If we denote the second product by ρ, the determinant of the function ζ is thus

$$= \pi^{m-2} \upsilon^{(m-1)(m-2)} \rho$$

Further, let r denote the function of the indeterminates x, l', l'', l''' etc. which becomes ρ under the substitutions $l' = \lambda', l'' = \lambda'', l''' = \lambda'''$ etc., and let R denote the function of x alone into which r passes under the substitutions $l' = L', l'' = L''$, $l''' = L'''$ etc. It is then clear that the determinant of the function z is

$$= p^{m-2} y^{(m-1)(m-2)} r$$

and that the determinant of the function Z is

$$= P^{m-2} Y^{(m-1)(m-2)} R$$

Since, by hypothesis, P is not $= 0$, it now remains to prove that R certainly cannot vanish identically.

15.

For this purpose we shall introduce one more indeterminate w and consider the product of all

$$(a+b-c-d)w + (a-c)(a-d)$$

with repetitions excluded. Since it involves a, b, c etc. symmetrically, it can be represented as a polynomial in the indeterminates $w, \lambda', \lambda'', \lambda'''$ etc. We shall denote this function by $f(w, \lambda', \lambda'', \lambda'''$ etc.). The number of the factors $(a+b-c-d)w + (a-c)(a-d)$ is

$$= \frac{1}{2} m(m-1)(m-2)(m-3)$$

from which we readily conclude that

$$f(0, \lambda', \lambda'', \lambda''' \text{ etc.}) = \pi^{(m-2)(m-3)}$$

and hence also

$$f(0, l', l'', l''' \text{ etc.}) = p^{(m-2)(m-3)}$$

as well as

$$f(0, L', L'', L''' \text{ etc.}) = P^{(m-2)(m-3)}$$

In general, the polynomial $f(w, L', L'', L''' \text{ etc.})$ will have degree

$$\frac{1}{2}m(m-1)(m-2)(m-3)$$

but in special cases it may have a lower degree, if some coefficients, starting from the highest power of w, happen to vanish. It is impossible, however, for the entire function to be identically $= 0$, since the equation just found shows that at least its last term does not vanish. Suppose that the highest term of the polynomial $f(w, L', L'', L''' \text{ etc.})$ having a nonvanishing coefficient is Nw^ν. If we substitute $w = x - a$, then $f(x - a, L', L'', L''' \text{ etc.})$ is clearly a polynomial in the indeterminates x, a, or, equivalently, a polynomial in x with coefficients depending on the indeterminate a. Its highest term is nevertheless Nx^ν, and thus has a fixed coefficient, independent of a, which is not $= 0$. Likewise, $f(x - b, L', L'', L''' \text{ etc.}), f(x - c, L', L'', L''' \text{ etc.})$ are polynomials in the indeterminate x, each with highest term Nx^ν, while the coefficients of the following terms depend respectively on b, c etc. Hence the product of the m factors

$$\begin{gathered} f(x - a, L', L'', L''' \text{ etc.}) \\ f(x - b, L', L'', L''' \text{ etc.}) \\ f(x - c, L', L'', L''' \text{ etc.}) \\ \text{etc.} \end{gathered}$$

is a polynomial in x whose highest term is $N^m x^{m\nu}$, while the coefficients of the following terms depend on the indeterminates a, b, c etc.

Next consider the product of these m factors:

$$\begin{gathered} f(x - a, l', l'', l''' \text{ etc.}) \\ f(x - b, l', l'', l''' \text{ etc.}) \\ f(x - c, l', l'', l''' \text{ etc.}) \\ \text{etc.} \end{gathered}$$

As a function of the indeterminates x, a, b, c etc., l', l'', l''' etc., symmetric with respect to a, b, c etc., it can be represented as a function of the indeterminates $x, \lambda', \lambda'', \lambda'''$ etc., l', l'', l''' etc., which we shall denote by

$$\varphi(x, \lambda', \lambda'', \lambda''' \text{ etc.},\ l', l'', l''' \text{ etc.})$$

Thus

$$\varphi(x, \lambda', \lambda'', \lambda''' \text{ etc.},\ \lambda', \lambda'', \lambda''' \text{ etc.})$$

is the product of the factors

$$\begin{gathered} f(x-a, \lambda', \lambda'', \lambda''' \text{ etc.}) \\ f(x-b, \lambda', \lambda'', \lambda''' \text{ etc.}) \\ f(x-c, \lambda', \lambda'', \lambda''' \text{ etc.}) \\ \text{etc.} \end{gathered}$$

and is therefore identically divisible by ρ, since it is easy to see that every factor of ρ occurs in one of those factors. We therefore set

$$\begin{aligned} &\varphi(x, \lambda', \lambda'', \lambda''' \text{ etc.},\ \lambda', \lambda'', \lambda''' \text{ etc.}) \\ &= \rho\psi(x, \lambda', \lambda'', \lambda''' \text{ etc.}) \end{aligned}$$

where the symbol ψ represents a polynomial. It follows readily that we also have identically

$$\begin{aligned} &\varphi(x, L', L'', L''' \text{ etc.},\ L', L'', L''' \text{ etc.}) \\ &= R\psi(x, L', L'', L''' \text{ etc.}) \end{aligned}$$

But we have proved above that the product of the factors

$$\begin{gathered} f(x-a, L', L'', L''' \text{ etc.}) \\ f(x-b, L', L'', L''' \text{ etc.}) \\ f(x-c, L', L'', L''' \text{ etc.}) \\ \text{etc.} \end{gathered}$$

which is $= \varphi(x, \lambda', \lambda'', \lambda''' \text{ etc.},\ L', L'', L''' \text{ etc.})$, has highest term $N^m x^{m\nu}$. The function $\varphi(x, L', L'', L''' \text{ etc.}, L', L'', L''' \text{ etc.})$ therefore has the same highest term, and is certainly not identically $= 0$. Hence R cannot be identically $= 0$ either, and neither can the determinant of the function Z. Q. E. D.

16.

THEOREM. *Let $\varphi(u,x)$*[*)] *denote a product of any number of factors in which the indeterminates u, x enter only linearly, that is, factors of the form*

$$\alpha+\beta u+\gamma x$$
$$\alpha'+\beta' u+\gamma' x$$
$$\alpha''+\beta'' u+\gamma'' x$$
$$\text{etc.}$$

and let w be another indeterminate. Then the function

$$\varphi(u+w\,.\,\tfrac{\mathrm{d}\varphi(u,x)}{\mathrm{d}x},\ x-w\,.\,\tfrac{\mathrm{d}\varphi(u,x)}{\mathrm{d}u})=\Omega$$

is identically divisible by $\varphi(u,x)$.

Proof. Setting

$$\begin{aligned}\varphi(u,x)&=(\alpha+\beta u+\gamma x)Q\\&=(\alpha'+\beta' u+\gamma' x)Q'\\&=(\alpha''+\beta'' u+\gamma'' x)Q''\\&\quad\text{etc.}\end{aligned}$$

we obtain for Q, Q', Q'' etc. polynomials in the indeterminates $u, x, \alpha, \beta, \gamma$, α', β', γ', $\alpha'', \beta'', \gamma''$ etc., and

$$\begin{aligned}\frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}x}&=\gamma Q+(\alpha+\beta u+\gamma x)\,.\,\frac{\mathrm{d}Q}{\mathrm{d}x}\\&=\gamma' Q'+(\alpha'+\beta' u+\gamma' x)\,.\,\frac{\mathrm{d}Q'}{\mathrm{d}x}\\&=\gamma'' Q''+(\alpha''+\beta'' u+\gamma'' x)\,.\,\frac{\mathrm{d}Q''}{\mathrm{d}x}\\&\quad\text{etc.}\end{aligned}$$

$$\begin{aligned}\frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}u}&=\beta Q+(\alpha+\beta u+\gamma x)\,.\,\frac{\mathrm{d}Q}{\mathrm{d}u}\\&=\beta' Q'+(\alpha'+\beta' u+\gamma' x)\,.\,\frac{\mathrm{d}Q'}{\mathrm{d}u}\\&=\beta'' Q''+(\alpha''+\beta'' u+\gamma'' x)\,.\,\frac{\mathrm{d}Q''}{\mathrm{d}u}\\&\quad\text{etc.}\end{aligned}$$

[*)] Every reader will see, even without our pointing it out, that the symbols introduced in the preceding article are confined to that article alone, and that the present meaning of the symbols φ, w must therefore not be confused with their former meaning.

When these values are substituted in the factors whose product is Ω, namely in

$$\alpha + \beta u + \gamma x + \beta w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}x} - \gamma w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}u}$$
$$\alpha' + \beta' u + \gamma' x + \beta' w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}x} - \gamma' w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}u}$$
$$\alpha'' + \beta'' u + \gamma'' x + \beta'' w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}x} - \gamma'' w \,.\, \frac{\mathrm{d}\varphi(u,x)}{\mathrm{d}u}$$
etc., respectively

these take the following values:

$$(\alpha + \beta u + \gamma x)(1 + \beta w \,.\, \tfrac{\mathrm{d}Q}{\mathrm{d}x} - \gamma w \,.\, \tfrac{\mathrm{d}Q}{\mathrm{d}u})$$
$$(\alpha' + \beta' u + \gamma' x)(1 + \beta' w \,.\, \tfrac{\mathrm{d}Q'}{\mathrm{d}x} - \gamma' w \,.\, \tfrac{\mathrm{d}Q'}{\mathrm{d}u})$$
$$(\alpha'' + \beta'' u + \gamma'' x)(1 + \beta'' w \,.\, \tfrac{\mathrm{d}Q''}{\mathrm{d}x} - \gamma'' w \,.\, \tfrac{\mathrm{d}Q''}{\mathrm{d}u})$$
etc.

Thus Ω is the product of $\varphi(u,x)$ and the factors

$$1 + \beta w \,.\, \frac{\mathrm{d}Q}{\mathrm{d}x} - \gamma w \,.\, \frac{\mathrm{d}Q}{\mathrm{d}u}$$
$$1 + \beta' w \,.\, \frac{\mathrm{d}Q'}{\mathrm{d}x} - \gamma' w \,.\, \frac{\mathrm{d}Q'}{\mathrm{d}u}$$
$$1 + \beta'' w \,.\, \frac{\mathrm{d}Q''}{\mathrm{d}x} - \gamma'' w \,.\, \frac{\mathrm{d}Q''}{\mathrm{d}u}$$

etc., that is, of $\varphi(u,x)$ and a polynomial in the indeterminates $u, x, w, \alpha, \beta, \gamma$, $\alpha', \beta', \gamma', \alpha'', \beta'', \gamma''$ etc. Q. E. D.

17.

The theorem of the preceding article clearly applies to the function ζ, which we shall now represent by

$$f(u, x, \lambda', \lambda'', \lambda''' \text{ etc.})$$

so that

$$f(u + w \,.\, \tfrac{\mathrm{d}\zeta}{\mathrm{d}x},\ x - w \,.\, \tfrac{\mathrm{d}\zeta}{\mathrm{d}u},\ \lambda', \lambda'', \lambda''' \text{ etc.})$$

is identically divisible by ζ. The quotient is a polynomial in the indeterminates u, x, w, a, b, c etc., symmetric with respect to a, b, c etc.; we shall represent it by

$$\psi(u, x, w, \lambda', \lambda'', \lambda''' \text{ etc.})$$

We conclude that we also have identically

$$f(u + w \,.\, \tfrac{\mathrm{d}z}{\mathrm{d}x},\ x - w \,.\, \tfrac{\mathrm{d}z}{\mathrm{d}u},\ l', l'', l''' \text{ etc.})$$
$$= z\psi(u, x, w, l', l'', l''' \text{ etc.})$$

and likewise

$$f(u + w\,.\,\tfrac{\mathrm{d}Z}{\mathrm{d}x},\ x - w\,.\,\tfrac{\mathrm{d}Z}{\mathrm{d}u},\ L', L'', L''' \text{ etc.})$$
$$= Z\psi(u, x, w, L', L'', L''' \text{ etc.})$$

Thus, if we represent the function Z simply by $F(u, x)$, so that

$$f(u, x, L', L'', L''' \text{ etc.}) = F(u, x)$$

holds, we have identically

$$F(u + w\,.\,\tfrac{\mathrm{d}Z}{\mathrm{d}x},\ x - w\,.\,\tfrac{\mathrm{d}Z}{\mathrm{d}u})$$
$$= Z\psi(u, x, w, L', L'', L''' \text{ etc.})$$

18.

Suppose, then, that fixed values of u, x, namely $u = U, x = X$, give

$$\frac{\mathrm{d}Z}{\mathrm{d}x} = X', \quad \frac{\mathrm{d}Z}{\mathrm{d}u} = U'$$

We have identically

$$F(U + wX',\ X - wU')$$
$$= F(U, X)\,.\,\psi(U, X, w, L', L'', L''' \text{ etc.})$$

Whenever U' does not vanish, we may set

$$w = \frac{X - x}{U'}$$

which gives

$$F(U + \tfrac{XX'}{U'} - \tfrac{X'x}{U'},\ x)$$
$$= F(U, X)\,.\,\psi(U, X, \tfrac{X-x}{U'}, L', L'', L''' \text{ etc.})$$

This can also be stated as follows:

If in the function Z we set $u = U + \frac{XX'}{U'} - \frac{X'x}{U'}$, it becomes

$$F(U, X)\,.\,\psi(U, X, \tfrac{X-x}{U'}, L', L'', L''' \text{ etc.})$$

19.

In the case where $P = 0$ does not hold, the determinant of the function Z is a function of the indeterminate x that does not vanish in itself. The number of fixed values of x for which this determinant can take the value 0 is therefore clearly finite. Thus infinitely many fixed values of x can be specified which give this determinant a value different from 0. Let X be such a value

of x, which we may also suppose to be *real*. The determinant of the function $F(u, X)$ is then not $= 0$. By Theorem II of Article 6, the functions

$$F(u, X) \quad \text{and} \quad \frac{\mathrm{d}F(u, X)}{\mathrm{d}u}$$

can therefore have no common divisor. Suppose further that there is a fixed value of u, namely U, either real or imaginary, that is, of the form $g + h\sqrt{-1}$, which makes $F(u, X) = 0$, so that $F(U, X) = 0$. Then $u - U$ is a factor of the polynomial $F(u, X)$, and the function $\frac{\mathrm{d}F(u,X)}{\mathrm{d}u}$ is consequently certainly not divisible by $u - U$. Suppose this function, $\frac{\mathrm{d}F(u,X)}{\mathrm{d}u}$, takes the value U' when $u = U$ is set. It certainly cannot be the case that $U' = 0$. Now U' is clearly the value of the partial derivative $\frac{\mathrm{d}Z}{\mathrm{d}u}$ for $u = U, x = X$. If we also denote by X' the value of the partial derivative $\frac{\mathrm{d}Z}{\mathrm{d}x}$ for the same values of u, x, the preceding article shows that the function Z vanishes identically under the substitution

$$u = U + \frac{XX'}{U'} - \frac{X'x}{U'}$$

and is therefore identically divisible by the factor

$$u + \frac{X'}{U'}x - (U + \frac{XX'}{U'})$$

Hence, on setting $u = xx$, we see that $F(xx, x)$ is divisible by

$$xx + \frac{X'}{U'}x - (U + \frac{XX'}{U'})$$

and thus takes the value 0 when x is taken to be a root of the equation

$$xx + \frac{X'}{U'}x - (U + \frac{XX'}{U'}) = 0$$

that is, when we set

$$x = \frac{-X' \pm \sqrt{(4UU'U' + 4XX'U' + X'X')}}{2U'}$$

These values are known to be either real or of the form $g + h\sqrt{-1}$.

It is now easy to prove that the function Y must also vanish for these same values of x. For $f(xx, x, \lambda', \lambda'', \lambda'''$ etc.) is clearly the product of all $(x - a)(x - b)$, with repetitions excluded, and thus is $= \upsilon^{m-1}$. It follows at once that

$$f(xx, x, l', l'', l''' \text{ etc.}) = y^{m-1}$$
$$f(xx, x, L', L'', L''' \text{ etc.}) = Y^{m-1}$$

or $F(xx, x) = Y^{m-1}$, whose fixed value therefore cannot vanish unless the value of Y vanishes at the same time.

20.

The preceding investigations have reduced the *solution* of the equation $Y = 0$, that is, the finding of a fixed value of x, either real or of the form $g + h\sqrt{-1}$, which satisfies it, to the solution of the equation $F(u, X) = 0$, provided that the determinant of the function Y is not $= 0$. It should be observed that, if all the coefficients in Y, that is, the numbers L', L'', L''' etc., are real quantities, all the coefficients in $F(u, X)$ are also real, provided that a real quantity is taken for X, as is allowed. The degree of the secondary equation $F(u, X) = 0$ is given by the number $\frac{1}{2}m(m-1)$. Whenever m is an even number of the form $2^\mu k$, with k denoting an undetermined odd number, the degree of the secondary equation is therefore given by a number of the form $2^{\mu-1}k$.

If the determinant of the function Y becomes $= 0$, Article 10 gives another polynomial $\mathfrak{Y}$ dividing it, whose determinant is not $= 0$ and whose degree is given by a number of the form $2^\nu k$, with either $\nu < \mu$ or $\nu = \mu$. Every solution of the equation $\mathfrak{Y} = 0$ also satisfies the equation $Y = 0$. The solution of the equation $\mathfrak{Y} = 0$ is again reduced to the solution of another equation whose degree is given by a number of the form $2^{\nu-1}k$.

We therefore conclude that, in general, the solution of any equation whose degree is an even number of the form $2^\mu k$ can be reduced to the solution of another equation whose degree is of the form $2^{\mu'}k$, with $\mu' < \mu$. Whenever this number is still even, that is, μ' is not $= 0$, the same method is applied again. We shall continue in this way until we reach an equation of odd degree. All its coefficients will be real if all the coefficients of the original equation were real. Such an equation of odd degree is known to have a solution, indeed a real root. Each of the preceding equations will therefore also have a solution, either through real roots or through roots of the form $g + h\sqrt{-1}$.

It has thus been established that any polynomial Y of the form $x^m - L'x^{m-1} + L''x^{m-2} -$ etc., where L', L'' etc. are fixed real quantities, contains a factor $x - A$, where A is a quantity either real or of the form $g + h\sqrt{-1}$. In the latter case it is easy to see that Y also takes the value 0 under the substitution $x = g - h\sqrt{-1}$, and is therefore divisible by $x - (g - h\sqrt{-1})$, hence also by the product $xx - 2gx + gg + hh$. Every polynomial Y therefore certainly contains a real factor of the first or second degree. Since the same holds again for the quotient, Y can clearly be resolved into real factors of the first or second degree. This was what the present paper set out to prove.

THIRD PROOF

OF THE THEOREM ON THE FACTORIZATION

OF POLYNOMIALS

INTO REAL FACTORS

SUPPLEMENT TO THE PRECEDING PAPER

B Y

CARL FRIEDRICH GAUSS

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THEOREM ON THE FACTORIZATION OF POLYNOMIALS INTO REAL FACTORS.

THIRD PROOF.

SUPPLEMENT TO THE PRECEDING PAPER.

After the preceding paper had already been printed, further reflection on the same subject led to a new proof of the theorem. Like the preceding proof, it is purely analytic, but it rests on entirely different principles and seems far preferable to it in simplicity. These few pages shall therefore be devoted to this *third* proof.

1.

Let the following function of the indeterminate x be given:

$$X = x^m + Ax^{m-1} + Bx^{m-2} + Cx^{m-3} + \text{etc.} + Lx + M$$

where the coefficients A, B, C etc. are fixed real quantities. Let r, φ be other indeterminates, and put

$$\begin{aligned} &r^m \cos m\varphi + Ar^{m-1} \cos(m-1)\varphi \\ &\qquad + Br^{m-2} \cos(m-2)\varphi \\ &\qquad + Cr^{m-3} \cos(m-3)\varphi + \text{etc.} + Lr\cos\varphi + M = t \\ &r^m \sin m\varphi + Ar^{m-1} \sin(m-1)\varphi \\ &\qquad + Br^{m-2} \sin(m-2)\varphi \\ &\qquad + Cr^{m-3} \sin(m-3)\varphi + \text{etc.} + Lr\sin\varphi = u \end{aligned}$$

$$
\begin{aligned}
&mr^m \cos m\varphi + (m-1)Ar^{m-1}\cos(m-1)\varphi \\
&\qquad + (m-2)Br^{m-2}\cos(m-2)\varphi \\
&\qquad + (m-3)Cr^{m-3}\cos(m-3)\varphi + \text{etc.} + Lr\cos\varphi = t' \\
&mr^m \sin m\varphi + (m-1)Ar^{m-1}\sin(m-1)\varphi \\
&\qquad + (m-2)Br^{m-2}\sin(m-2)\varphi \\
&\qquad + (m-3)Cr^{m-3}\sin(m-3)\varphi + \text{etc.} + Lr\sin\varphi = u'
\end{aligned}
$$

$$
\begin{aligned}
&mmr^m \cos m\varphi + (m-1)^2 Ar^{m-1}\cos(m-1)\varphi \\
&\qquad + (m-2)^2 Br^{m-2}\cos(m-2)\varphi \\
&\qquad + (m-3)^2 Cr^{m-3}\cos(m-3)\varphi + \text{etc.} + Lr\cos\varphi = t'' \\
&mmr^m \sin m\varphi + (m-1)^2 Ar^{m-1}\sin(m-1)\varphi \\
&\qquad + (m-2)^2 Br^{m-2}\sin(m-2)\varphi \\
&\qquad + (m-3)^2 Cr^{m-3}\sin(m-3)\varphi + \text{etc.} + Lr\sin\varphi = u''
\end{aligned}
$$

$$
\frac{(tt+uu)(tt''+uu'') + (tu'-ut')^2 - (tt'+uu')^2}{r(tt+uu)^2} = y
$$

The factor r can evidently be cancelled from the denominator of the last formula, since t', u', t'', u'' are divisible by it. Finally, let R be a fixed positive quantity, arbitrary except that it must exceed the largest of the quantities

$$
mA\sqrt{2}, \quad \sqrt{(mB\sqrt{2})}, \quad \sqrt[3]{(mC\sqrt{2})}, \quad \sqrt[4]{(mD\sqrt{2})} \quad \text{etc.}
$$

disregarding the signs of the quantities A, B, C etc., that is, replacing any negative quantities among them by positive ones. With these preparations, I assert that $tt' + uu'$ certainly takes a positive value when we put $r = R$, whatever (real) value is assigned to φ.

Proof. Put

$$\begin{aligned}
&R^m \cos 45^\circ + AR^{m-1}\cos(45^\circ+\varphi)\\
&\quad + BR^{m-2}\cos(45^\circ+2\varphi)\\
&+ CR^{m-3}\cos(45^\circ+3\varphi) + \text{etc.}\\
&\quad + LR\cos(45^\circ+(m-1)\varphi) + M\cos(45^\circ+m\varphi) = T\\
&R^m \sin 45^\circ + AR^{m-1}\sin(45^\circ+\varphi)\\
&\quad + BR^{m-2}\sin(45^\circ+2\varphi)\\
&+ CR^{m-3}\sin(45^\circ+3\varphi) + \text{etc.}\\
&\quad + LR\sin(45^\circ+(m-1)\varphi) + M\sin(45^\circ+m\varphi) = U\\
&mR^m \cos 45^\circ + (m-1)AR^{m-1}\cos(45^\circ+\varphi)\\
&\quad + (m-2)BR^{m-2}\cos(45^\circ+2\varphi)\\
&+ (m-3)CR^{m-3}\cos(45^\circ+3\varphi) + \text{etc.}\\
&\quad + LR\cos(45^\circ+(m-1)\varphi) = T'\\
&mR^m \sin 45^\circ + (m-1)AR^{m-1}\sin(45^\circ+\varphi)\\
&\quad + (m-2)BR^{m-2}\sin(45^\circ+2\varphi)\\
&+ (m-3)CR^{m-3}\sin(45^\circ+3\varphi) + \text{etc.}\\
&\quad + LR\sin(45^\circ+(m-1)\varphi) = U'
\end{aligned}$$

It is evident that

I. T is composed of the parts

$$\begin{aligned}
&\frac{R^{m-1}}{m\sqrt{2}}[R + mA\sqrt{2}\,.\,\cos(45^\circ+\varphi)]\\
&+ \frac{R^{m-2}}{m\sqrt{2}}[RR + mB\sqrt{2}\,.\,\cos(45^\circ+2\varphi)]\\
&+ \frac{R^{m-3}}{m\sqrt{2}}[R^3 + mC\sqrt{2}\,.\,\cos(45^\circ+3\varphi)]\\
&+ \frac{R^{m-4}}{m\sqrt{2}}[R^4 + mD\sqrt{2}\,.\,\cos(45^\circ+4\varphi)]\\
&+ \text{etc.}
\end{aligned}$$

each of which is readily seen to be positive for any fixed real value of φ. Hence T necessarily takes a positive value. Similarly, it is proved that U, T', U' are also positive, so that $TT' + UU'$ is necessarily a positive quantity as well.

II. For $r = R$, the functions t, u, t', u' become, respectively,

$$T\cos(45^\circ + m\varphi) + U\sin(45^\circ + m\varphi)$$
$$T\sin(45^\circ + m\varphi) - U\cos(45^\circ + m\varphi)$$
$$T'\cos(45^\circ + m\varphi) + U'\sin(45^\circ + m\varphi)$$
$$T'\sin(45^\circ + m\varphi) - U'\cos(45^\circ + m\varphi)$$

as is readily proved by expansion. Hence the value of the function $tt' + uu'$, for $r = R$, is found to be $= TT' + UU'$, and is therefore a positive quantity. Q. E. D.

The same formulas also show that the value of the function $tt + uu$, for $r = R$, is $TT + UU$, and is therefore positive. We conclude that for no value of r greater than each of $mA\sqrt{2}, \sqrt{(mB\sqrt{2})}, \sqrt[3]{(mC\sqrt{2})}$ etc. can $t = 0, u = 0$ hold simultaneously.

2.

THEOREM. *Within the limits $r = 0$ and $r = R$, and $\varphi = 0$ and $\varphi = 360^\circ$, there certainly exist values of the indeterminates r, φ for which $t = 0$ and $u = 0$ hold simultaneously.*

Proof. Suppose the theorem is false. Evidently, the value of $tt + uu$ must then be a positive quantity for all values of the indeterminates within the stated limits, and the value of y must therefore always be finite. Consider the double integral

$$\iint y\,\mathrm{d}r\,\mathrm{d}\varphi$$

extended from $r = 0$ to $r = R$, and from $\varphi = 0$ to $\varphi = 360^\circ$, which therefore takes a finite, fully determined value. This value, which we shall denote by Ω, must be the same whether we integrate first with respect to φ and then with respect to r, or in the reverse order. But, treating r as constant, we have the *indefinite integral*

$$\int y\,\mathrm{d}\varphi = \frac{tu' - ut'}{r(tt + uu)}$$

as is readily confirmed by differentiation with respect to φ. No constant is to be added if we suppose the integral to begin at $\varphi = 0$, since for $\varphi = 0$ we have $\frac{tu'-ut'}{r(tt+uu)} = 0$. Thus, since $\frac{tu'-ut'}{r(tt+uu)}$ evidently also vanishes for $\varphi = 360^\circ$, the integral $\int y\,\mathrm{d}\varphi$ from $\varphi = 0$ to $\varphi = 360^\circ$ is $= 0$, while r remains indeterminate. It follows that $\Omega = 0$.

Likewise, treating φ as constant, we have the indefinite integral

$$\int y \, \mathrm{d}r = \frac{tt' + uu'}{tt + uu}$$

as is just as readily confirmed by differentiation with respect to r. Here too no constant is to be added when the integral begins at $r = 0$. Thus, by what was proved in the preceding article, the integral extended from $r = 0$ to $r = R$ is $= \frac{TT'+UU'}{TT+UU}$, and so, by the theorem of that article, is always a positive quantity for every real value of φ. Hence Ω too, that is, the value of the integral

$$\int \frac{TT' + UU'}{TT + UU} \, \mathrm{d}\varphi$$

from $\varphi = 0$ to $\varphi = 360°$, is necessarily a positive quantity*). This is absurd, since we previously found the same quantity to be $= 0$. The supposition therefore cannot stand, and the truth of the theorem is established.

3.

The function X becomes $t + u\sqrt{-1}$ under the substitution $x = r(\cos\varphi + \sin\varphi \,.\sqrt{-1})$, and $t - u\sqrt{-1}$ under the substitution $x = r(\cos\varphi - \sin\varphi \,.\sqrt{-1})$. Thus, if for fixed values of r, φ, say $r = g, \varphi = G$, we have $t = 0, u = 0$ simultaneously (such values were proved to exist in the preceding article), X takes the value 0 under each substitution

$$x = g(\cos G + \sin G \,.\sqrt{-1}), \quad x = g(\cos G - \sin G \,.\sqrt{-1})$$

and is therefore identically divisible by

$$x - g(\cos G + \sin G \,.\sqrt{-1}), \quad \text{and also by} \quad x - g(\cos G - \sin G \,.\sqrt{-1})$$

Whenever neither $\sin G = 0$ nor $g = 0$ holds, these divisors are unequal, and X is therefore also divisible by their product

$$xx - 2g\cos G \,.\, x + gg$$

But whenever either $\sin G = 0$, and hence $\cos G = \pm 1$, or $g = 0$ holds, those factors are identical, namely $= x \mp g$. Thus the function X certainly contains a real divisor of the second or first degree. Since the same conclusion applies again to the quotient, X can be completely resolved into such factors. Q. E. D.

*) As is already evident in itself. Moreover, the indefinite integral is readily found to be $= m\varphi + 45° - \text{arc.tang}\frac{U}{T}$, and it can be proved *by another argument* (for it is not yet evident in itself which of the infinitely many values belonging to the multivalued function arc. tang. $\frac{U}{T}$ should be chosen for $\varphi = 360°$) that its value, extended to $\varphi = 360°$, must be put equal to $= m \times 360°$, or $= 2m\pi$. But this is not necessary for our purpose.

4.

Although we have already fully accomplished the task before us, it will not be superfluous to add a few further remarks on the reasoning of Article 2. From the supposition that t and u do not vanish simultaneously for any values of the indeterminates r, φ within the limits stated there, we arrived at an unavoidable contradiction, from which we concluded that the supposition was false. This contradiction must therefore disappear if there really are values of r, φ for which t and u simultaneously become $= 0$. To clarify this further, we observe that for such values $tt + uu = 0$, so that y itself becomes infinite. We may therefore no longer treat the double integral $\int\int y \, \mathrm{d}r \, \mathrm{d}\varphi$ as a quantity with a definite value. Generally speaking, if ξ, η, ζ denote indeterminate coordinates of points in space, the integral $\int\int y \, \mathrm{d}r \, \mathrm{d}\varphi$ represents the volume of the solid contained between the five planes whose equations are

$$\xi = 0, \quad \eta = 0, \quad \zeta = 0, \quad \xi = R, \quad \eta = 360^\circ$$

and the surface whose equation is $\zeta = y$, taking as negative those parts in which the coordinates ζ are negative. But it is tacitly understood here that the sixth surface is *continuous*. When this condition fails as y becomes infinite, it may well happen that this concept has no meaning. In such a case there can be no question of summing the integral $\int\int y \, \mathrm{d}r \, \mathrm{d}\varphi$, and it is therefore no wonder that analytic operations applied by blind calculation to empty concepts lead to absurdities.

The integration $\int y \, \mathrm{d}\varphi = \frac{tu' - ut'}{r(tt+uu)}$ is a true integration, that is, a summation, only insofar as y is finite everywhere between the limits over which it extends; it is absurd if y becomes infinite anywhere between those limits. If we evaluate an integral such as $\int \eta \, \mathrm{d}\xi$, which generally represents the area between the axis of abscissas and the curve whose ordinate is $= \eta$ for the abscissa ξ, by the usual rules, without regard to continuity, we very often become entangled in contradictions. For example, putting $\eta = \frac{1}{\xi\xi}$, analysis gives the integral $= C - \frac{1}{\xi}$, which correctly determines the area as long as the curve remains continuous. But continuity is interrupted at $\xi = 0$, and if someone inappropriately asks for the magnitude of the area from a negative abscissa to a positive one, the formula will give the absurd answer that it is negative. We shall discuss more fully on another occasion what these and similar paradoxes of analysis mean.

One more observation may be added here. When questions that may become absurd in certain cases are posed *without restriction*, analysis very often provides for itself by making the answer partly indeterminate. Thus, for the value of the integral $\int\int y \, \mathrm{d}r \, \mathrm{d}\varphi$ extended from $r = e$ to $r = f$, and from $\varphi = E$ to $\varphi = F$, if the value of $\frac{u}{t}$

$$\begin{aligned} &\text{for } r = e,\ \varphi = E \quad \text{is denoted by} \quad \theta \\ &\qquad r = e,\ \varphi = F \quad . \ . \ . \ . \ . \ . \ . \quad \theta' \\ &\qquad r = f,\ \varphi = E \quad . \ . \ . \ . \ . \ . \ . \quad \theta'' \\ &\qquad r = f,\ \varphi = F \quad . \ . \ . \ . \ . \ . \ . \quad \theta''' \end{aligned}$$

analytic operations readily give

$$\text{Arc. tang}\,\theta - \text{Arc. tang}\,\theta' - \text{Arc. tang}\,\theta'' + \text{Arc. tang}\,\theta'''$$

In fact, the integral can have a definite value only when y always remains finite within the stated limits. This value is certainly contained in the formula given, but is not yet completely determined by it, since Arc. tang. is a multivalued function. It will be necessary to decide separately, by other considerations (which are not difficult), precisely which values of the function are to be used in a given case. On the other hand, whenever y becomes infinite anywhere within the stated limits, the question of the value of the integral $\int\int y \, \mathrm{d}r \, \mathrm{d}\varphi$ is absurd. If, despite this, you insist on forcing an answer from analysis, it will give now one answer, now another, according to the different methods; nevertheless, every one of these answers will be contained in the general formula given above.

PROOF

Of an

ALGEBRAIC THEOREM.

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Proof
OF AN ALGEBRAIC THEOREM.

The subject of this paper is the theorem of Cartes, usually named after Harriot, concerning the relation between the number of positive and negative roots of an algebraic equation and the number of alternations and continuations in the signs of its coefficients. The proofs of this theorem attempted by various writers lack the clarity, brevity and comprehensive generality that one may rightly demand of so elementary a subject. A new treatment of it therefore seems not superfluous.

Let X be a polynomial in x of degree m, arranged in descending powers of x. We assume, without loss of generality, that its leading term is x^m and that the constant term, which is independent of x, is not missing. Only the terms actually present are to be written down; any missing terms are therefore not to be inserted with coefficient 0.

If the coefficients are not all positive, they will exhibit one or more changes of sign. Let $-Nx^n$ be the first negative term, $+Px^p$ the first positive term after it, $-Qx^q$ the first negative term after that, and so on. Thus m, n, p, q etc. are decreasing integers, and N, P, Q etc. are positive, and X is represented as follows:

$$X = x^m + + \,..\, - Nx^n - - \,..\, + Px^p + + \,..\, - Qx^q - \text{etc.}$$

Multiply X by the linear factor $x - \alpha$, where α is assumed positive. It is easy to see that in the product x^{n+1} has a negative coefficient, x^{p+1} a positive one, x^{q+1} a negative one, and so on, so that the product has the form

$$X(x-\alpha) = x^{m+1} \ldots - N'x^{n+1} \ldots + P'x^{p+1} \,..\, - Q'x^{q+1} \ldots$$

where N', P', Q' etc. are positive. The signs between the displayed terms remain undetermined. But it is clear that from the leading term to the power x^{n+1} there is at least one change of sign, to x^{p+1} at least two, to x^{q+1} at least three, and so on. If the last change of sign in X occurs at the term $\pm Ux^u$, and the coefficient of x^{u+1} in $X(x-\alpha)$ is denoted by $\pm U'$, then U' is positive, and by the term $\pm U'x^{u+1}$ at least as many changes of sign have occurred as there are in X. But the last term of $X(x-\alpha)$ has the sign $\mp$, so at least one further change of sign must have occurred by then. We conclude that $X(x-\alpha)$ has at least one more change of sign than X.

Now let X be the product of all the linear factors corresponding to the negative and imaginary roots of an equation $y = 0$. Thus, if α, β, γ etc. are the positive roots of the same equation,

$$y = X(x-\alpha)(x-\beta)(x-\gamma)\ldots$$

By the preceding proposition, $X(x-\alpha)$ has at least one more change of sign than X, $X(x-\alpha)(x-\beta)$ at least two more, $X(x-\alpha)(x-\beta)(x-\gamma)$ at least three more, and so on. Hence, even if X has no change of sign at all, y has at least as many changes of sign as there are positive roots. It is evident that when the equation has neither negative nor imaginary roots we must put $X = 1$, and this conclusion remains valid.

Let y become y' when the signs of the coefficients of the powers x^{m-1}, x^{m-3}, x^{m-5} etc. are reversed. All the roots of the equation $y' = 0$ are then the opposites of the roots of $y = 0$. Hence y' has at least as many changes of sign as the equation $y = 0$ has negative roots.

We therefore have the following theorem:

The equation $y = 0$ cannot have more positive roots than there are changes of sign in y, or more negative roots than there are changes of sign in y'.

This formulation of the theorem seems the most suitable, since it combines the greatest simplicity with the most comprehensive generality, and all the forms of the theorem that hold only under special conditions follow from it of themselves.

To read the bound on the number of negative roots directly from the signs of the coefficients of y, we must distinguish *direct* sign changes and sign continuations, between terms whose exponents of x differ by one, from those *interrupted* by missing terms. Evidently, every direct sign change in y', and every sign change interrupted by an even number of missing terms, corresponds to a sign continuation of the same kind in y, whereas a sign change in y' interrupted by an odd number of missing terms remains a sign change of the same kind in y. The second part of the theorem can therefore also be expressed as follows:

The number of negative roots of the equation $y = 0$ cannot exceed the number of direct sign continuations and those interrupted by an even number of missing terms, added to the number of sign changes in y interrupted by an odd number of missing terms.

If no term is missing from y, the number of negative roots does not exceed the number of sign continuations.

If A denotes the number of direct sign changes, and B the number of direct sign continuations in y, then, when no term is missing, $A + B = m$, the number of all the roots. Since these signs tell us only that the number of positive roots cannot exceed A, and the number of negative roots cannot exceed B, it remains undecided whether there are imaginary roots, or how many. But if we know from another source that the equation has no imaginary. roots, then A must necessarily equal the number of positive roots, and B the number of negative roots.

The situation is different when terms are missing from y. To see clearly what can be concluded from this about the imaginary roots, let a denote the number of sign changes interrupted by an even number of missing terms, and c the number interrupted by an odd number; let b and d denote, respectively, the numbers of sign continuations in y interrupted by an even and an odd number of missing terms. It is easy to see that $m - A - B - a - b - c - d$ equals the total number of missing terms, which we shall denote by e. Now, by our theorem, the number of positive roots is at most $A + a + c$, and the number of negative roots at most $B + b + c$, so the number of all the real roots is at most

$$A + B + a + b + 2c = m + c - d - e$$

Hence the number of imaginary roots must be at least $e - c + d$.

Thus, count all the missing terms, except that in each gap with an odd number of missing terms, count one fewer at a sign change and one more at a sign continuation. The number obtained cannot exceed the number of imaginary roots.

CONTRIBUTIONS TO THE THEORY

OF

ALGEBRAIC EQUATIONS

BY

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CONTRIBUTIONS TO THE THEORY OF ALGEBRAIC EQUATIONS.

This memoir treats two different subjects concerning algebraic equations. First, I present the proof of the fundamental theorem of the theory of algebraic equations that I gave fifty years ago, in a modified form and with substantial additions. The second part is devoted to a special treatment of algebraic equations with three terms and contains methods for easily determining not only their real but also their imaginary roots.

FIRST DIVISION.

The memoir published in 1799, *A New Proof of the Theorem That Every Polynomial in One Variable Can Be Resolved into Real Factors of the First or Second Degree*, had two aims: first, to show that all the proofs previously attempted of this most important theorem of the theory of algebraic equations were insufficient and illusory, and second, to give a new, fully rigorous proof. There is no need to return to the first subject. I myself later followed the new proof given there with two others, and a fourth was first given by CAUCHY. These four proofs all rest on as many different foundations. They agree, however, in that each initially proves only the existence of *one* factor of the function in question. This certainly does not detract from the rigour of the proofs. For it is clear that removing this one factor from the given function leaves a similar function of lower degree, to which the theorem can be applied again, and that repeating the procedure must ultimately give a complete factorization of the original function into factors of the specified kind. Nevertheless, every proof undoubtedly attains a higher degree of completeness if it can be shown to make the existence of all the factors directly evident. I already indicated in that memoir, Article 23, that the first proof does this, without developing the point further there. This shall now be supplied, and I take the opportunity to repeat the main points of the entire proof in a modified form that, I believe, offers greater clarity. As for the outward formulation of the theorem itself, the form used in 1799, that the function $x^n + Ax^{n-1} + Bx^{n-2} + \text{etc.}$ can be resolved into real factors of the first or second degree, was chosen then because all use of imaginary quantities was to be avoided. Now that the concept of complex quantities is familiar to everyone, it seems more appropriate to abandon that form and state the theorem by saying that the function can be resolved into n *linear* factors. The constant parts of these factors need not be real quantities; any complex values must

also be allowed for them. With this formulation the theorem itself gains in generality, since the restriction to real values need not be imposed on the coefficients A, B etc. either; any values remain admissible for them.

1.

We therefore consider the function of the indeterminate x

$$x^n + Ax^{n-1} + Bx^{n-2} + \text{etc.} + Mx + N = X$$

where $A, B \ldots . M, N$ denote fixed real or imaginary coefficients.

The relation between the roots of the equation $X = 0$ and the linear factors of X is known from elementary algebra. Namely, if the substitution $x = p$ satisfies that equation, then $x - p$ is a factor of X. If there are n different ways of satisfying it, namely $x = p$, $x = p'$, $x = p''$ etc., then the product $(x - p)\ (x - p')(x - p'') \ldots .$ is identical to X. Under special circumstances, however, a solution such as $x = p$ may also give the factor $(x-p)^2$, or $(x-p)^3$, or some higher power in X. In these cases the root p is counted as occurring twice, three times, and so on.

Thus, if only a proof that the function X certainly admits *one* linear factor is required, it suffices to establish the existence of any root of the equation $X = 0$. But if the complete factorization of the function into linear factors is to be proved all at once, it must be shown that the equation $X = 0$ can be satisfied either by n unequal values of x, or by a smaller number of unequal solutions, some of which have the characteristics of equal roots counted more than once, in such a way that counting all the unequal and equal roots gives the total $= n$.

2.

The entire domain of complex quantities, in which the values of x satisfying the equation $X = 0$ are to be sought, is an infinite domain of two dimensions. For if such a value is put in the form $x = t + iu$, where i always denotes the imaginary unit $\sqrt{-1}$, all real values from $-\infty$ to $+\infty$ are admissible for t and u. We must first delimit within this infinite domain a finite one outside which no root of the given equation $X = 0$ can possibly lie. This can be done in more than one way; the following seems best suited to our purpose.

Instead of the form $t + iu$, use

$$x = r(\cos\rho + i\sin\rho)$$

To cover the entire infinite domain of complex quantities, r must then range over all positive values from 0 to $+\infty$, and ρ from 0 to 360^0, or, what is the same, from any initial value to a final value greater by 360^0.

To obtain a bound for r beyond which no value can correspond to a root of the equation $X = 0$, I first put the coefficients of the individual terms of X into a form similar to that of x, namely

$$\begin{aligned} A &= a(\cos\alpha + i\sin\alpha) \\ B &= b(\cos\beta + i\sin\beta) \\ C &= c(\cos\gamma + i\sin\gamma) \quad \text{etc.} \end{aligned}$$

Here a, b, c are to denote fixed positive quantities, except that one or another of them may also be $= 0$. I then consider the equation

$$r^n - \sqrt{2}\,.(ar^{n-1} + br^{n-2} + cr^{n-3} + \text{etc.}) = 0$$

which, as is readily seen, has a positive root, and indeed, by HARRIOT's theorem, only one such root. Let R be this root. It is then evident that for every positive value of r greater than R, the value of $r^n - \sqrt{2}\,.(ar^{n-1} + br^{n-2} + cr^{n-3} + \text{etc.})$ is positive, and the same also holds for the function

$$nr^n - \sqrt{2}\,.((n-1)ar^{n-1} + (n-2)br^{n-2} + (n-3)cr^{n-3} + \text{etc.})$$

since it exceeds n times the first function by

$$\sqrt{2}\,.(ar^{n-1} + 2br^{n-2} + 3cr^{n-3} + \text{etc.})$$

and thus by a positive difference.

3.

I now assert that the quantity R can serve as a bound for the values of r of the kind required in the preceding article. The proposition is to be proved as follows.

In general I put $X = T + iU$, where T and U of course denote real quantities. In fact,

$$\begin{aligned} T = {} & r^n \cos n\rho + ar^{n-1}\cos((n-1)\rho + \alpha) + br^{n-2}\cos((n-2)\rho + \beta) \\ & + cr^{n-3}\cos((n-3)\rho + \gamma) + \text{etc.} \end{aligned}$$

$$\begin{aligned} U = {} & r^n \sin n\rho + ar^{n-1}\sin((n-1)\rho + \alpha) + br^{n-2}\sin((n-2)\rho + \beta) \\ & + cr^{n-3}\sin((n-3)\rho + \gamma) + \text{etc.} \end{aligned}$$

It is easy to see that when any positive value greater than R is chosen for r, T necessarily has the same sign as $\cos n\rho$ whenever the absolute value of this cosine is not less than $\sqrt{\frac{1}{2}}$. For this it is enough to put T into the form

$$\begin{aligned}\pm T = \sqrt{\tfrac{1}{2}}.r^n - ar^{n-1} - br^{n-2} - cr^{n-3} - \text{etc.}\\ + [\pm\cos n\rho - \sqrt{\tfrac{1}{2}}]r^n\\ + [1 \pm \cos((n-1)\rho + \alpha)]ar^{n-1}\\ + [1 \pm \cos((n-2)\rho + \beta)]br^{n-2}\\ + [1 \pm \cos((n-3)\rho + \gamma)]cr^{n-3}\\ + \text{etc.}\end{aligned}$$

The upper signs are to apply when $\cos n\rho$ is positive, and the lower signs when it is negative. The first part of the expression on the right is positive by the proposition of the preceding article, while all the remaining parts are at least nonnegative. In much the same way, by writing U in place of T in the above formula and replacing cosines by sines throughout, it follows that under the same assumption about r, U always has the same sign as $\sin n\rho$ whenever the absolute value of this sine is not less than $\sqrt{\frac{1}{2}}$. Thus, in every case, at least one of the two quantities T, U has a predetermined positive or negative sign. Consequently, for no value of ρ can the function satisfy $X = 0$. Q. E. D.

4.

To clarify further the behaviour of T and U with regard to their signs and changes of sign, for a fixed value of r exceeding R, let ρ run through all values between two limits differing by 360^0. Instead of 0 and 360^0, however, taking for brevity

$$\frac{45^0}{n} = \omega$$

the limits $-\omega$ and $(8n-1)\omega$ are to be chosen. I divide the entire range into $4n$ equal parts, the first extending from $-\omega$ to ω, the second from ω to 3ω, the third from 3ω to 5ω, and so on. First the values of the derivatives $\frac{\mathrm{d}T}{\mathrm{d}\rho}$, $\frac{\mathrm{d}U}{\mathrm{d}\rho}$ must also be considered. They are given by

$$\begin{aligned}\tfrac{\mathrm{d}T}{\mathrm{d}\rho} &= -nr^n \sin n\rho - (n-1)ar^{n-1}\sin\big((n-1)\rho+\alpha\big)\\ &\quad -(n-2)br^{n-2}\sin\big((n-2)\rho+\beta\big)\big)\\ &\quad -(n-3)cr^{n-3}\sin\big((n-3)\rho+\gamma\big) - \text{etc.}\\ \tfrac{\mathrm{d}U}{\mathrm{d}\rho} &= nr^n\cos n\rho + (n-1)ar^{n-1}\cos\big((n-1)\rho+\alpha\big)\\ &\quad +(n-2)br^{n-2}\cos\big((n-2)\rho+\beta\big)\big)\\ &\quad +(n-3)cr^{n-3}\cos\big((n-3)\rho+\gamma\big) + \text{etc.}\end{aligned}$$

By reasoning similar to that of the preceding article, and using the proposition at the end of Article 2, it is easy to see from these formulas that $\frac{\mathrm{d}T}{\mathrm{d}\rho}$ always has the opposite sign to $\sin n\rho$ whenever the absolute value of this sine is not less than $\sqrt{\frac{1}{2}}$, whereas $\frac{\mathrm{d}U}{\mathrm{d}\rho}$ always has the same sign as $\cos n\rho$ whenever the absolute value of this cosine is not less than $\sqrt{\frac{1}{2}}$. The following conclusions result.

In the first interval, that is, from $\rho = -\omega$ to $\rho = +\omega$, T is always positive, whereas U is negative at the initial value and positive at the final value. It is therefore certainly $= 0$ once in between, and *only* once, since $\frac{\mathrm{d}U}{\mathrm{d}\rho}$ is positive throughout the interval.

In the second interval, U is always positive, while T is positive at the beginning and negative at the end. In between, $T = 0$ holds once, and only once, since $\frac{\mathrm{d}T}{\mathrm{d}\rho}$ is negative throughout the interval.

In the third interval, T is always negative, while U undergoes a change of sign, so that $U = 0$ holds *once*.

In the fourth interval, U is always negative, and T is $= 0$ *once*.

In the following intervals these relations repeat in the same order, so that the fifth corresponds to the first, the sixth to the second, and so on.

5.

From the order of the positive and negative values of T and U discussed in the preceding article, which holds for every value of r greater than R[*)], it can now be inferred that within the domain of smaller values of r there must be certain crossings in these arrangements which contain the essence of the theorem we are to prove. I shall present the proof in a form drawn from the geometry of position, because this gives it the greatest clarity and simplicity. At bottom, however, the actual content of the whole argument belongs to a higher domain of the general abstract theory of magnitudes, independent of spatial considerations, whose subject is combinations of magnitudes connected through continuity. This domain has so far been little cultivated, and one cannot move within it without a language borrowed from spatial images.

6.

The entire domain of complex quantities is represented by an unbounded plane, in which each point whose coordinates relative to two mutually perpendicular axes are t, u is regarded as corresponding to the complex quantity $x = t + iu$. If this complex quantity is put in the form $x = r(\cos\rho + i\sin\rho)$,

[*)] It is easy to show that the same order remains valid even for the value $r = R$ itself, with only the restriction that, in quite special cases, a transition value of ρ (that is, one at which T is zero or $U = 0$) may coincide with one of the quantities $-\omega$, ω, 3ω, 5ω, etc., whereas for all greater values of r, every transition value of ρ must lie *between* two of these quantities. I shall not dwell on this, however, since for our purpose it is enough to have established that this order holds from some value of r onwards.

then r, ρ denote the polar coordinates of the corresponding point. The collection of all complex quantities for which r has one fixed value is therefore represented by a circle whose radius is this value and whose centre is the origin of coordinates. I shall denote by K that circle for which r exceeds R by a difference chosen at will, and by (1), (2), (3) . . . . (2n) those points on it corresponding to the values of ρ lying respectively between ω and 3ω, between 5ω and 7ω, between 9ω and 11ω, and so on, up to between $(8n-3)\omega$ and $(8n-1)\omega$, for which, by Article 4, $T = 0$. Notice that U will be positive at the points (1), (3), (5), etc., and negative at the points (2), (4), (6), etc.

7.

The collection of points in our plane for which T is positive forms connected regions, as is evident on considering that, in a continuous passage from one point to another, T changes continuously. Likewise, all the points for which T is negative form connected regions. Between the regions of the first kind and those of the second lie points at which $T = 0$; by the nature of the function T, these points cannot themselves form areas, but only lines bounding regions of the one kind on one side and regions of the other kind on the other.

The space outside K contains n regions of the first kind alternating with just as many of the second kind, each extending continuously to infinity from a portion of the circumference K. At the same time, it is clear that each of these regions continues beyond the circumference into the interior, and that the following cases can occur as regards its further shape.

1) The region in question, beginning at a portion of K, ends in isolation within the disk. Its peripheral boundary then consists of only two connected pieces, one of which is part of K, while the other lies within the circle. In the accompanying figure, which concerns an equation of the fifth degree and in which the signs of T are written in the various regions, three of the regions with positive T fall into this case. One has the boundary lines 10.1 and 1.11.10, the second has 4.5 and 5.12.4, and the third 6.7 and 7.13.6. There are two similar regions with negative T.

2) The region simply crosses the disk so as to form one connected region with another entering at a different place. The entire peripheral boundary will then consist of four pieces, two belonging to the circumference K and the other two to the interior. In our figure this case occurs in the region bounded by 2.3; 3.0.8; 8.9; 9.11.2.

3) The region branches once or several times within the circle, so as to form one connected region with two or more others entering at other places. Its entire peripheral boundary will then consist of six, eight, or more pieces in an even number, belonging alternately to the circumference and to the interior. In our figure this occurs in a region whose boundary is formed by the six pieces 3.4; 4.12.5; 5.6; 6.13.7; 7.8; 8.0.3, but in which T is negative.

8.

A complete enumeration of all conceivable shapes of the regions entering the circle would require further modifications of the cases given. For example, if such a region divides into two branches which join again in the interior, this could be included in the first or second case of the preceding article, according as, after they join, the region ends in the interior or continues (without dividing again) to another place on the circumference. Its shape would be modified only by enclosing an island that does not belong to it. Moreover, it would not be difficult to prove rigorously that, owing to the special nature of the function T, modifications of this kind are *not possible at all*. For our purpose, however, this is unnecessary, since only the succession of pieces of the *outer* boundary of each region in question matters (that is, of those in which T is positive).

We have already observed that the number of these pieces is always even (two in the first case of the preceding article, four in the second, six or more in the third), one belonging to the circumference K and one to the interior, alternately. Further, it is clear that if this outer boundary is traversed always in the same direction, here chosen as that in which the numbers of the points of K increase (thus, for example, in our figure, so that the region always lies to the right of the boundary), the initial and final points of a piece belonging to the circumference will be designated respectively by an even number and the odd number greater by one. Thus the initial and final points of every piece passing through the interior are always designated respectively by an odd and an even number.

It is therefore established that each of the n lines which enter the interior at a point of K designated by an odd number, and on which $T = 0$ everywhere, passes continuously through this space in a completely definite way[*)], until it emerges at another place designated by an even number. Now, as already observed above (at the end of Article 6), the value of U is positive at its initial point and negative at its final point. By continuity of the change in value, $U = 0$ must therefore hold at an intermediate point. This point then represents a root of the equation $X = 0$; and since the number of such lines $= n$, n roots of that equation always result in this way.

[*)] That it always has a completely definite course rests on its being part of the outer boundary of a region in which T has a definite sign. I have chosen the positive sign, which in itself is entirely arbitrary. Understood in this way, for example, the line entering at 1 continues through 11 to 10. As part of the boundary of a region in which T is negative, the line 1.11 would have to be continued to 2. If, on the other hand, one speaks only of a line on which $T = 0$, without regarding it as part of the boundary of a particular region, 11.9 could more readily be considered the natural continuation of 1.11. The point of view chosen here distinguishes my present procedure from that of 1799, and contributes substantially to simplifying the proof.

9.

If the lines in question pass through the circle without meeting one another, it is clear that the n roots obtained in this way are necessarily distinct. Such an unobstructed passage occurs in our figure for the lines from 3 to 8, from 5 to 4, and from 7 to 6, to which belong the roots represented by the points 0, 12, 13. If, however, two or more such lines have a point in common, it is not thereby necessary, but is nevertheless possible, that this point is also the one at which $U = 0$. In that case two or more roots coincide in one, or, as is usually expressed, are equal to one another. In our figure the lines 1.10 and 9.2 meet at the point 11, and at this point $U = 0$ also holds. Thus, besides the three distinct roots already listed, the equation has two equal roots.

10.

It remains only to show that, if the point P representing a root $= p$ lies on two or more lines $T = 0$ at once, the square of $x - p$, or the higher power corresponding to the number of these concurrent lines, will be contained in X as a factor. The proof rests on the following propositions.

Introduce another indeterminate quantity z in place of x by putting $x = z + p$. Let this substitution turn X into Z, so that Z is a function of z of the same degree as X in x, but without a constant term. Arrange it in ascending powers of z, and let its lowest nonvanishing term be

$$= Kz^m \quad \text{and} \quad Z = Kz^m(1 + \zeta)$$

where ζ will have the form $Lz + L'zz + L''z^3 + \text{etc.} + \frac{1}{K}z^{n-m}$. Finally put

$$z = s(\cos\psi + i\sin\psi)$$

The real and imaginary parts of z express the position of any indeterminate point of the plane as rectangular coordinates, and the quantities s, ψ as polar coordinates, relative to the point P, in exactly the same way as the parts of x and the quantities r, φ designate the relative position with respect to the original origin. The combination of a fixed value of s with all values of ψ over a range of 360^0 thus represents the points of a circumference with centre P and radius $= s$.

If one now puts $K = k(\cos\kappa + i\sin\kappa)$, and consequently

$$Kz^m = ks^m(\cos(m\psi + \kappa) + i\sin(m\psi + \kappa))$$

then, for an infinitely small s, the quantity ζ, which is of at least the same order as s, may be neglected beside 1, and it is therefore permissible to put

$$T = ks^m\cos(m\psi + \kappa)$$

It follows that, as ψ increases by 360^0, the sign of T is positive on m pieces of the circumference and negative on just as many alternating with them;

or that T takes the value $= 0$ at $2m$ points, namely for $\psi = \frac{1}{m}(\kappa - 90^0)$, $\frac{1}{m}(\kappa + 90^0)$, $\frac{1}{m}(\kappa + 270^0)$, etc. Thus a total of $2m$ lines on which $T = 0$ issue from P. Or, if they are joined in pairs so that each line at which, as ψ increases, the sign passes from $-$ to $+$ is regarded, together with the next one at which the opposite transition occurs, as the boundary of a region with positive T, then in all m such boundary lines meet at P.

On the other hand, it is clear that, just as Z is identically divisible by z^m and by no higher power of z, X will contain the factor $(x-p)^m$, but no higher power of $x - p$. Thus, whenever p denotes any root of the equation $X = 0$, the exponent of the highest power of $x - p$ by which X is divisible is equal to the number of boundary lines of regions with positive T meeting *at* P, or, what is the same, to the number of such regions meeting *at* P.

Moreover, it is easy to give the proof a form entirely independent of any introduction of infinitely small quantities, and quite analogous to the chain of reasoning in Articles 3 and 4. Namely, one can establish a value of s for which, as for every smaller value, the entire cycle of all values of ψ presents the same alternating succession of m pieces with positive T and just as many with a negative value. The positive root of the equation

$$0 = m\sqrt{\frac{1}{2}} - (m+1)ls - (m+2)l'ss - (m+3)l''s^3 - \text{etc.}$$

has this property. Here l, l', l'', etc. denote the positive square roots of the norms of the complex quantities L, L', L'', etc.; alternatively, put

$$\begin{aligned} L &= l\,(\cos\lambda + i\sin\lambda) \\ L' &= l'(\cos\lambda' + i\sin\lambda') \\ L'' &= l''(\cos\lambda'' + i\sin\lambda'') \text{ etc.} \end{aligned}$$

I believe, however, that I may pass over the very easy development of this proposition here.

Finally, it may be observed that the proof in the paper of 1799 required the consideration of *two* systems of lines, one containing the lines where $T = 0$, the other those where $U = 0$, whereas in our present procedure the consideration of one system has sufficed. For this I chose the system of boundary lines of the regions with positive T, but consideration of the boundary lines of the regions with positive (or negative) U could have served the same purpose equally well.

SECOND DIVISION.

11.

Various methods can be used to determine numerically the roots of algebraic equations consisting of only three terms. Here these methods admit an elegance and convenience far beyond the laborious operations unavoidable for equations of a less simple form. Such methods therefore deserve a separate exposition, especially since equations of this form occur often enough.

This applies first to the expansion of the roots into infinite series. In fact, *every* root of an equation with three terms, whether real or imaginary, can be expressed by a convergent series with a simple law of progression. For several reasons, however, I shall exclude *this* mode of solution entirely from the present discussion. I only observe here that the rate of convergence depends on the relation between the coefficients: the closer this relation comes to that for which the equation has two equal roots, the slower the convergence; and in this limiting case itself it is slower than that of any decreasing geometric progression. However noteworthy these series may be from a general theoretical point of view, in practice one will always prefer the *indirect* methods to be developed in the following articles, except when the series converge very rapidly.

12.

To find the *real* roots I use my auxiliary table for logarithms of sums and differences, first printed in 1810, or, when greater accuracy is required than logarithms of five digits can give, the similar but extended table of Matthiessen. I have already published a few special applications of this procedure: to the solution of quadratic equations, in the twentieth edition of Vega's logarithmic handbook, published in 1840; and to the solution of the cubic equation used to determine the true anomaly in parabolic motion, in No. 474 of the Astronomische Nachrichten. In the latter place the general applicability of the procedure to all algebraic equations with three terms was also pointed out. Although the treatment of this entirely elementary subject presents no difficulty, the considerable variety of cases will make a clear division of them and a collection of rules ready for use welcome enough to justify a few pages.

Instead of these auxiliary logarithmic tables one can also use the ordinary tables of trigonometric logarithms. The former, however, are generally more convenient for the present purpose and give twice the accuracy of the latter. I would therefore restrict the use of trigonometric tables for this task to the rare case in which one wishes to exceed even the accuracy attainable with logarithms of seven digits, and can use for this purpose the well-known logarithms of ten digits in Vlacq's or Vega's Thesaurus. Moreover, when

auxiliary logarithms are used, twice as many cases must be distinguished as when trigonometric logarithms are used. This must not be regarded as a disadvantage: once the complete general classification is available, it is easy to assign each particular case to its class, and the actual indirect task is much easier when the whole class has only half the extent. But for precisely this reason the general classification is shorter and more convenient to present for solution by trigonometric logarithms. I shall therefore give it first, since the classification for the other mode of solution then follows of itself.

13.

The method is applied directly only to determining the *positive* roots of a given equation. The negative roots are obtained by applying the same procedure to the equation resulting from the given one on introducing the quantity opposite to the original unknown.

I put the equation in the form

$$x^{m+n} \pm ex^m \pm f = 0$$

where m, n, e, f denote given *positive* quantities. Depending on the combination of signs, this form actually comprises four different cases; but the first, in which the upper signs are chosen both times, is excluded, since the equation

$$x^{m+n} + ex^m + f = 0$$

clearly cannot have a positive root. Moreover, it is permissible to assume that m and n (which are understood to be integers, although the applicability of the method itself is independent of this) have no common divisor, since every other case is easily reduced to this one. Finally, for brevity I shall write

$$\frac{f^n}{e^{m+n}} = \lambda$$

First Form.

$$x^{m+n} + ex^m - f = 0$$

Introduce an angle θ, always to be taken in the first quadrant, so that

$$\frac{x^{m+n}}{f} = \sin\theta^2, \quad \frac{ex^m}{f} = \cos\theta^2$$

and hence (I)

$$x^{m+n} = f\sin\theta^2, \quad x^m = \frac{f\cos\theta^2}{e}, \quad x^n = e\text{tang}\theta^2$$

Eliminating x gives the equation

$$\lambda = \frac{\sin\theta^{2m}}{\cos\theta^{2m+2n}}$$

from which θ must be determined. It is easy to see that the right-hand side of this equation, regarded as a function of an indeterminate quantity θ, increases from 0 to ∞ as θ passes through all values from 0 to 90^0, and that there is therefore one, and only one, value of θ satisfying the equation. Once it has been found, x is obtained from one of the formulas I. Notice that $\theta = 45^0$ when $\lambda = 2^n$, and consequently θ is to be sought in the first octant when λ is smaller, and in the second when λ is greater than 2^n.

Second Form.

$$x^{m+n} - ex^m - f = 0$$

Here put

$$fx^{-m-n} = \sin\theta^2, \quad ex^{-n} = \cos\theta^2$$

or (I)

$$x^{m+n} = \frac{f}{\sin\theta^2}, \quad x^n = \frac{e}{\cos\theta^2}, \quad x^m = \frac{f\,\mathrm{cotang}\theta^2}{e}$$

Thus θ is to be determined from the equation

$$\lambda = \frac{\sin\theta^{2n}}{\cos\theta^{2m+2n}}$$

which can be done in one and only one way. The value of x is then found from one of the equations I. The angle θ lies in the first or second octant according as λ is smaller or greater than 2^m.

Third Form.

$$x^{m+n} - ex^m + f = 0$$

Here put

$$\frac{x^n}{e} = \sin\theta^2, \quad \frac{fx^{-m}}{e} = \cos\theta^2$$

or (I)

$$x^{m+n} = f\,\mathrm{tang}\theta^2, \quad x^m = \frac{f}{e\cos\theta^2}, \quad x^n = e\sin\theta^2$$

One of these formulas will serve to determine x as soon as the value of θ has been found. This is obtained by solving the equation

$$\lambda = \cos\theta^{2n}\sin\theta^{2m}$$

Since the right-hand side of this equation, regarded as a function of an indeterminate quantity θ, vanishes both for $\theta = 0$ and for $\theta = 90^0$, there must be a greatest value between them. Since the differential of the logarithm of this function $= (2m\text{cotg}\theta - 2n\text{tang}\theta)\,\mathrm{d}\theta$, the greatest value occurs for $\theta = \theta^*$, on putting $\sqrt{\frac{m}{n}} = \text{tang}\theta^*$. The function therefore increases from 0 to its greatest value, evidently

$$= \frac{m^m n^n}{(m+n)^{m+n}}$$

and then decreases to 0, as θ increases from 0 to θ^* and thence to 90^0. Thus the maximum value is always greater than the value for $\theta = 45^0$, that is, greater than $\frac{1}{2^{m+n}}$, except when $m = n$, in which case $\frac{1}{2^{m+n}}$ itself is the maximum value.

It follows that, according as λ is greater than

$$\frac{m^m n^n}{(m+n)^{m+n}}$$

or smaller, the equation $\lambda = \cos\theta^{2n} \sin\theta^{2m}$ cannot be satisfied at all, or can be satisfied by two distinct values of θ. In the former case the equation $x^{m+n} - ex^m + f = 0$ has no (positive) root; in the latter it has two. In the special case where

$$\lambda = \frac{m^m n^n}{(m+n)^{m+n}}$$

the two solutions coincide and the equation has two equal roots, for which any of the three formulas may be used at will:

$$x^{m+n} = \frac{fm}{n}, \quad x^m = \frac{f(m+n)}{en}, \quad x^n = \frac{em}{m+n}$$

As for the octants in which the values of θ fall when two solutions actually exist, it is easy to see that, if λ is greater than $\frac{1}{2^{m+n}}$, both values of θ lie in the same octant as θ^*, namely in the first or second according as m is smaller or greater than n. If, however, λ is smaller than $\frac{1}{2^{m+n}}$, one value of θ is to be sought in the first octant and the other in the second. In the special case where $\lambda = \frac{1}{2^{m+n}}$, 45^0 itself is one value of θ, and the other lies in the same octant as θ^*.

One may also note the consequence, easily obtained from this analysis of all three forms, that our equation (insofar as we assume that m and n have no common divisor) cannot have more than three real roots, as is also known on other grounds.

14.

The preceding rules can now easily be recast as rules for the use of auxiliary logarithms, since these $A = \log a$. $B = \log b$, $C = \log c$, can be regarded as the logarithms of the squares of the tangents, cosecants and secants of angles increasing from 45^0 to 90^0, or, equivalently, as the logarithms of the squares of the cotangents, secants and cosecants of angles decreasing from 45^0 to 0. Thus

$$a = \text{tang}\theta^2, \quad \frac{1}{b} = \sin\theta^2, \quad \frac{1}{c} = \cos\theta^2$$

for values of θ in the second octant, or

$$\frac{1}{a} = \text{tang}\theta^2, \quad \frac{1}{c} = \sin\theta^2, \quad \frac{1}{b} = \cos\theta^2$$

for values of θ in the first octant.

I combine the complete rules in the following scheme, where, as above,

$$\lambda = \frac{f^n}{e^{m+n}}$$

is put.

First Form.

$$x^{m+n} + ex^m - f = 0$$

First Case. $\lambda > 2^n$

$$\lambda = a^{m+n}b^n = a^m c^n = \frac{c^{m+n}}{b^m}$$

$$x^{m+n} = \frac{f}{b}, \quad x^m = \frac{f}{ec}, \quad x^n = ea$$

Second Case. $\lambda < 2^n$

$$\lambda = \frac{b^n}{a^m} = \frac{c^n}{a^{m+n}} = \frac{b^{m+n}}{c^m}$$

$$x^{m+n} = \frac{f}{c}, \quad x^m = \frac{f}{eb}, \quad x^n = \frac{e}{a}$$

Second Form.

$$x^{m+n} - ex^m - f = 0$$

First Case. $\lambda > 2^m$

$$\lambda = a^{m+n}b^m = a^n c^m = \frac{c^{m+n}}{b^n}$$

$$x^{m+n} = fb, \quad x^m = \frac{f}{ea}, \quad x^n = ec$$

Second Case. $\lambda < 2^m$

$$\lambda = \frac{b^m}{a^n} = \frac{c^m}{a^{m+n}} = \frac{b^{m+n}}{c^n}$$

$$x^{m+n} = fc, \quad x^m = \frac{fa}{e}, \quad x^n = eb$$

Third Form.

$$x^{m+n} - ex^m + f = 0$$

First Case. $\frac{1}{\lambda} < \frac{(m+n)^{m+n}}{m^m n^n}$

No solution at all.

Second Case. $\frac{1}{\lambda} = \frac{(m+n)^{m+n}}{m^m n^n}$

Two equal roots, for whose determination one of the equations

$$x^{m+n} = \frac{fm}{n}, \quad x^m = \frac{f(m+n)}{en}, \quad x^n = \frac{em}{m+n}$$

is used.

Third Case. $\frac{1}{\lambda}$ greater than $\frac{(m+n)^{m+n}}{m^m n^n}$ but not greater than 2^{m+n}, and at the same time m greater than n.

Two roots, for which

$$\frac{1}{\lambda} = a^n b^{m+n} = \frac{c^{m+n}}{a^m} = b^m c^n$$

$$x^{m+n} = fa, \quad x^m = \frac{fc}{e}, \quad x^n = \frac{e}{b}$$

Fourth Case. The same bounds for $\frac{1}{\lambda}$ as in the third case, but m smaller than n.

Two roots, for which

$$\frac{1}{\lambda} = a^m b^{m+n} = \frac{c^{m+n}}{a^n} = b^n c^m$$

$$x^{m+n} = \frac{f}{a}, \quad x^m = \frac{fb}{e}, \quad x^n = \frac{e}{c}$$

Fifth Case. $\frac{1}{\lambda}$ greater than 2^{m+n}.

Two roots, one determined by the formulas of the third case and the other by those of the fourth. It may also be noted that in the third case the value of a corresponding to one root is smaller than $\frac{m}{n}$, while that corresponding to the other is greater than $\frac{m}{n}$; in the fourth case the two values of a stand in a similar relation to $\frac{n}{m}$.

15.

The following remarks on the use of these rules should be added.

To determine each root, two operations must be performed: first, derive from λ the corresponding value of a (and hence at the same time that of b or c); then calculate from this the value of x. For each of these two operations there is a choice of three formulas; in most cases I prefer those listed first. In all these calculations one deals with the logarithms alone, and never with the quantities λ, a, b, c themselves. The first operation is indirect and therefore generally involves several successive approximations. It will be convenient to begin with tables having fewer digits. MATTHIESSEN's table has seven decimal places, as is well known; mine has five; ENCKE and URSIN have had it printed with four digits. If, at the start of the work, one has no knowledge even of an initial rough approximation, it may be advantageous to have before one, on a separate small sheet, a still shorter extract from the tables with only three digits, for example:

A	B	A	B	A	B
0	0,301	1,0	0,041	2,0	0,004
0,1	0,254	1,1	0,033	2,1	0,003
0,2	0,212	1,2	0,027	2,2	0,003
0,3	0,176	1,3	0,021	2,3	0,002
0,4	0,146	1,4	0,017	2,4	0,002
0,5	0,119	1,5	0,014	2,5	0,001
0,6	0,097	1,6	0,011	2,9	0,001
0,7	0,079	1,7	0,009	3,0	0,000
0,8	0,064	1,8	0,007		
0,9	0,051	1,9	0,005		
1,0	0,041	2,0	0,004		

16.

As an example, take the equation

$$x^7 + 28x^4 - 480 = 0$$

for which

$$\lambda = \frac{6750}{823543}, \quad \log \frac{1}{\lambda} = 2{,}0863825$$

The equation has the first form and hence *one* positive root; since λ is smaller than 8, it belongs to the second case. The first operation consists in satisfying the equation $\log \frac{1}{\lambda} = 4A - 3B$, or, if the calculation is begun with three decimal places, the equation

$$2{,}086 = 4A - 3B$$

Even a quick glance at the table above shows that A is to be sought between 0,5 and 0,6. For

A	$4A - 3B$	Error
0,5	1,643	−0,443
0,6	2,109	+0,023

which suggests the more accurate value 0,595. A new calculation with the tables of five decimal places, thus putting $\log \frac{1}{\lambda} = 2{,}08638$, gives

A	$4A - 3B$	Error
0,595	2,08501	−0,00137
0,596	2,08961	+0,00323

from which the still more accurate value 0,5953 is found. Finally, for seven decimal places,

A	$4A - 3B$	Error
0,5952	2,0859279	−0,0004546
0,5953	2,0863885	+0,0000060

The value $A = 0{,}5953$ therefore still requires the correction $-\frac{60}{4606}$ units in the fourth decimal place. I retain it in this form, since, if x is to be determined by the first formula

$$x^7 = \frac{f}{c}$$

all that is needed is to find the corresponding value of C. This is obtained by adding to the value listed beside $A = 0{,}5953$, namely $C = 0{,}6935705$, the correction $-\frac{60}{4606} \times 798$, the latter being understood in units of the seventh decimal place. Thus

$$\begin{aligned} C &= 0{,}6935695 \\ \log f &= 2{,}6812412 \\ \hline 7 \log x &= 1{,}9876717 \\ \log x &= 0{,}2839531 \\ x &= 1{,}9228841 \end{aligned}$$

To find the negative roots, put $x = -y$ and seek the positive roots of the equation

$$y^7 - 28y^4 + 480 = 0$$

This has the third form. Since $\frac{1}{\lambda} = \frac{823543}{6750}$ is greater than $\frac{7^7}{3^3 4^4} = \frac{823543}{6912}$ but smaller than $2^7 = 128$, and at the same time m is greater than n, the third case applies: there are two roots, to find which the equation

$$2{,}0863825 = 3A + 7B$$

must be satisfied. From the closing remark of Article 14 it is known that one value of A must be smaller and the other greater than $\log \frac{4}{3} = 0{,}12494$.

Bounds for the values of A also follow at once from the table above with logarithms of three digits, which gives:

A	$3A+7B$	Error
0,0	2,107	+0,021
0,1	2,078	−0,008
0,2	2,084	−0,002
0,3	2,132	+0,046

If logarithms of four digits are to be used first for a more accurate determination, then for the first solution one initially has

A	$3A+7B$	Error
0,05	2,0869	+0,0005
0,06	2,0847	−0,0017

The tables of five digits then give

0,052	2,08667	+0,00029
0,053	2,08638	0

and finally those of seven digits

0,0529	2,0863943	+0,0000118
0,0530	2,0863660	−0,0000165

Hence

$$\begin{aligned} A &= 0{,}0529417 \\ \log f &= 2{,}6812412 \\ 7\log y &= 2{,}7341829 \\ \log y &= 0{,}3905976 \\ -y = x &= -2{,}4580892 \end{aligned}$$

For the second solution, the calculation, carried out in a similar way, is as follows:

A	$3A+7B$	Error
0,19	2,0843	−0,0021
0,20	2,0868	+0,0004

0,197	2,08627	−0,00011
0,198	2,08654	+0,00016

0,1975	2,0863805	−0,0000020
0,1976	2,0864082	+0,0000257

$$\begin{aligned} A &= 0{,}1975072 \\ \log f &= 2.6812412 \\ \hline 7\log y &= 2{,}8787484 \\ \log y &= 0{,}4112498 \\ x &= -2{,}5778036 \end{aligned}$$

The equation used here as our example was deliberately chosen so that two of its roots differ little. In such a case, as already observed in Article 11 above, the series are of little use because of their very slow convergence. At least a weak analogy can also be seen in the indirect solution: the successive approximations progress a little more slowly for the two negative roots, which are precisely the two that differ little, than for the positive root. There is, however, an essential difference: the very slow convergence of the series occurs for all the roots, whereas in the indirect procedure the slower approximation, noticeable even then only to a small degree, occurs solely for the two roots that differ little.

17.

The procedure that must be used to determine the imaginary roots is quite different from that taught in the preceding articles. In general, finding imaginary roots by an indirect method is far more difficult than finding real roots, because the former must be sought in an infinite region of two dimensions, the latter in an infinity of only one dimension. Precisely for this reason, a very broad special case in which that difficulty can be avoided and the question brought into the same domain as the search for real roots deserves a separate treatment. Equations with three terms provide such a case.

Since the method can be applied with equal ease whether the coefficients of the equation are real or imaginary, I take at once the general form of the equation as my starting point:

$$\mathrm{X} = x^{m+n} + e(\cos\varepsilon + i\sin\varepsilon)x^m + f(\cos\varphi + i\sin\varphi) = 0$$

where e and f denote positive quantities. For a real coefficient, positive or negative, the corresponding angle (ε or φ) is then either 0 or 180^0. Here too the assumption that m and n have no common divisor can be retained without loss of generality. An imaginary root $x = t + iu$ satisfying the equation is put in the form $r(\cos\rho + i\sin\rho)$. For our present purpose it is more advantageous to omit the usual condition that r be positive and require instead that ρ always be taken between the limits 0 and 180^0. When both coefficients of the equation are real, the range of values of ρ can be further reduced by half. For, as is known, the imaginary roots of such an equation belong together in pairs, such as $t + iu$ and $t - iu$. It is then evident that, for one root of each pair, the value of ρ falls between 0 and 90^0, and only these need be determined by the indirect procedure: the other follows of itself by exchanging ρ for $180^0 - \rho$ and r for $-r$.

18.

The essence of the method is to set up an equation containing only ρ, without r. To obtain it, divide the equation X = 0 by its first term, putting it in the form

$$1 + e(\cos\varepsilon + i\sin\varepsilon)x^{-n} + f(\cos\varphi + i\sin\varphi)x^{-m-n} = 0$$

or

$$\begin{aligned} 1 &+ er^{-n}(\cos(n\rho - \varepsilon) - i\sin(n\rho - \varepsilon)) \\ &+ fr^{-m-n}(\cos[(m+n)\rho - \varphi] - i\sin[(m+n)\rho - \varphi]) = 0 \end{aligned}$$

Since the imaginary parts here must cancel, one has (I)

$$r^m = -\frac{f\sin((m+n)\rho - \varphi)}{e\sin(n\rho - \varepsilon)}$$

Similarly, dividing the equation X = 0 by its second or third term and observing that in both cases the imaginary parts of the new equations must cancel gives the equations

$$\left.\begin{aligned} r^{m+n} &= \frac{f\sin(m\rho + \varepsilon - \varphi)}{\sin(n\rho - \varepsilon)} \\ r^n &= -\frac{e\sin(m\rho + \varepsilon - \varphi)}{\sin((m+n)\rho - \varphi)} \end{aligned}\right\} \quad \text{(I)}$$

Each of the three equations (I) can evidently also be derived by combining the other two. But eliminating r from a combination of two gives (II)

$$\lambda = (-1)^{m+n}\frac{\sin(m\rho + \varepsilon - \varphi)^m \sin(n\rho - \varepsilon)^n}{\sin((m+n)\rho - \varphi)^{m+n}}$$

where, for brevity, as above,

$$\frac{f^n}{e^{m+n}} = \lambda$$

is put. The various values of ρ must be determined from this equation. The value of r corresponding to each value of ρ is then found from one of the equations (I), most conveniently from the second as regards its absolute value. If $m + n$ is even, however, one of the other two equations must also be used to decide the sign.

19.

The indirect solution of equation II can always be carried out easily. Taking the following observations into account will help further.

1) The values of ρ lie between 0 and 180^0. When the coefficients of the given equation are real, only half the number need be sought individually, namely those lying between 0 and 90^0.

2) In either case, first divide the relevant interval into the subdivisions formed by changes of sign in the values of the function of ρ on the right-hand side of equation II. The transition values of ρ can evidently occur only where one of the angles $m\rho+\varepsilon-\varphi,\ n\rho-\varepsilon,\ (m+n)\rho-\varphi$ is divisible by 180^0, so that the function itself becomes either 0 or infinite. Those subdivisions in which the value of the function becomes negative are then excluded of themselves from further investigation.

3) Unless approximate values of ρ can already be obtained in other ways, the indirect search through the appropriate intervals will be made much easier by carrying out the first trials with shortened tables having few digits, as illustrated by the examples of Article 16. In some cases it may be convenient to begin by using only logarithms of sines with three digits, written on a small sheet, say at intervals of one degree.

20.

For further illustration, the calculation of the imaginary roots of the equation treated above,

$$x^7 + 28x^4 - 480 = 0$$

may serve as an example. In the notation of Article 17 we first have, as above, $m = 4,\ n = 3,\ e = 28,\ f = 480$, and further $\varepsilon = 0,\ \varphi = 180^0$. The formulas I of Article 18 therefore become

$$r^4 = \frac{480 \sin 7\rho}{28 \sin 3\rho}$$
$$r^7 = -\frac{480 \sin 4\rho}{\sin 3\rho}$$
$$r^3 = -\frac{28 \sin 4\rho}{\sin 7\rho}$$

and formula II becomes

$$\frac{1}{\lambda} = \frac{823543}{6750} = \frac{\sin 7\rho^7}{\sin 3\rho^3 \sin 4\rho^4}$$

From this equation two values lying between 0 and 90^0 must be determined for ρ, since the equation X = 0, besides its three real roots already found, has two pairs of imaginary roots belonging together. Within these limits, $\sin 7\rho = 0$ at three values, namely $\rho = 25\frac{5}{7}$ degrees, $51\frac{3}{7}$ degrees and $77\frac{1}{7}$ degrees; $\sin 7\rho^7$ changes sign each time. Likewise, $\sin 3\rho = 0$ once, for $\rho = 60^0$, with a change of sign in $\sin 3\rho^3$. Finally, $\sin 4\rho = 0$ once, for $\rho = 45^0$, but without a change of sign in $\sin 4\rho^4$. Taking into account also that the value of $\frac{\sin 7\rho^7}{\sin 3\rho^3 \sin 4\rho^4}$ for $\rho = 0$ is to be set equal to the limiting value $\frac{7^7}{3^3 4^4}$, the behaviour of this

function in the six subdivisions of the interval from 0 to 90^0 is summarized in the following table:

$\rho = 0$ degrees	$\frac{823543}{6912}$
	$+$
$25\frac{5}{7}$	0
	$-$
45	∞
	$-$
$51\frac{3}{7}$	0
	$+$
60	∞.
	$-$
$77\frac{1}{7}$	0
	$+$
90^0	∞

It follows that a value of ρ satisfying formula II must lie in each of the fourth and sixth intervals. Nothing further is needed for our purpose, since it is already known from the outset that there are only two such values. I put equation II in the form

$$7 \log\sin 7\rho - 3 \log\sin 3\rho - 4 \log\sin 4\rho = S = 2{,}0863825$$

The following scheme shows how the value lying between $51\frac{3}{7}$ and 60 degrees is found by successive approximation with tables of 3, 4, 5, 7 digits:

ρ	S	Error
57^0	1,527	$-0{,}559$
58	2,354	$+0{,}268$

$57^0 40'$	2,0624	$-0{,}0240$
57 50	2,2057	$+0{,}1193$

$57^0 41'$	2,07658	$-0{,}00980$
57 42	2,09074	$+0{,}00436$

$57^0 41' 41''$	2,0862962	$-0{,}0000863$
57 41 42	2,0865320	$+0{,}0001495$

This gives $\rho = 57^0 41' 41'' 366$, and then, from the second of the formulas I,

$\log\sin 4\rho$	$=$	$9{,}8891425\, n$
Compl. $\log\sin 3\rho$	$=$	$0{,}9193523$
$\log(-480)$	$=$	$2{,}6812412\, n$
$7 \log r$	$=$	$3{,}4897360$
$\log r$	$=$	$0{,}4985337$

and hence

$$x = +1{,}6843159 + 2{,}6637914\,i$$

as well as the other root belonging to it,

$$x = +1{,}6843159 - 2{,}6637914\,i$$

The other value lying between $77\frac{1}{7}$ and 90^0 for ρ is found, by using tables of three decimal places, to lie between 86^0 and 87^0. The calculation, in the same form as in the preceding case, is:

ρ	S	Error
86^0	1,885	−0,201
87	2,533	+0,447

$86^0 10'$	1,9907	−0,0957
86 20	2,0946	+0,0082

86 19	2,08409	−0,00229
86 20	2,09447	+0,00809

$86^0 19' 13''$	2,0863229	−0,0000596
86 19 14	2,0864970	+0,0001145

$$\rho = 86^0 19' 13'' 342$$

$\log \sin 4\rho$	=	$9{,}4049540\,n$
Compl. $\log \sin 3\rho$	=	$0{,}0081108\,n$
$\log(-480)$	=	$2{,}6812412\,n$
$7 \log r$	=	$2{,}0943060\,n$
$\log r$	=	$0{,}2991866\,n$

If one prefers r to be positive, it is only necessary at the same time to give ρ the value increased by 180^0, namely $266^0 19' 13'' 342$. The root itself is

$$x = -0{,}1278113 - 1{,}9874234\,i$$

and the other root belonging to it differs only in the sign of its imaginary part.

All the roots of the equation $x^7 + 28x^4 - 480 = 0$ are therefore

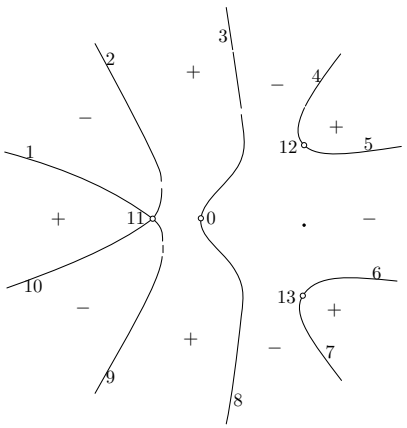
$$\begin{aligned}
&+1{,}9228841\\
&-2{,}4580892\\
&-2{,}5778036\\
&+1{,}6843159 + 2{,}6637914\,i\\
&+1{,}6843159 - 2{,}6637914\,i\\
&-0{,}1278113 + 1{,}9874234\,i\\
&-0{,}1278113 - 1{,}9874234\,i
\end{aligned}$$

The sum of the roots, +0,0000005, agrees with the true value 0 as accurately as could be expected from the use of logarithms of seven digits. In the other form one has

$\log r$	ρ
0,2839531	0
0,3905976	180^0
0,4112498	180
0,4985337	57 41′41″366
0,4985337	302 18 18,634
0,2991866	93 40 46,658
0,2991866	266 19 13,342

The sum of the logarithms of the values of r is found to be = 2,6812411, likewise agreeing with satisfactory accuracy with the logarithm of 480.

It is hardly necessary to recall that the calculations set out in this article and in Article 16 are intended only to illustrate the main stages of the work, and are by no means to prescribe the form of the minor mechanics of the operations. Those with more experience in calculation will usually prefer fewer intermediate stages than were used in those examples. In general, anyone with some experience in such work will readily arrange the details in the form best suited to the circumstances of each case and to their own individual habits. This is not the place to enter further into such details.



Gauss, Works, Volume III, page 102